CS11212 **- Spring** 2023

Data Structures & Introduction to **Algorithms**

Data Structures

Stacks & Queues

Ibrahim Albluwi

Stacks (a new ADT)

Stack

```
void push(const T& val)
    T pop()
    T top() const
```

add `val` to the collection of elements

remove the most recently added element

return the most recently added element

Stacks (a new ADT)

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```
void push(const T& val)
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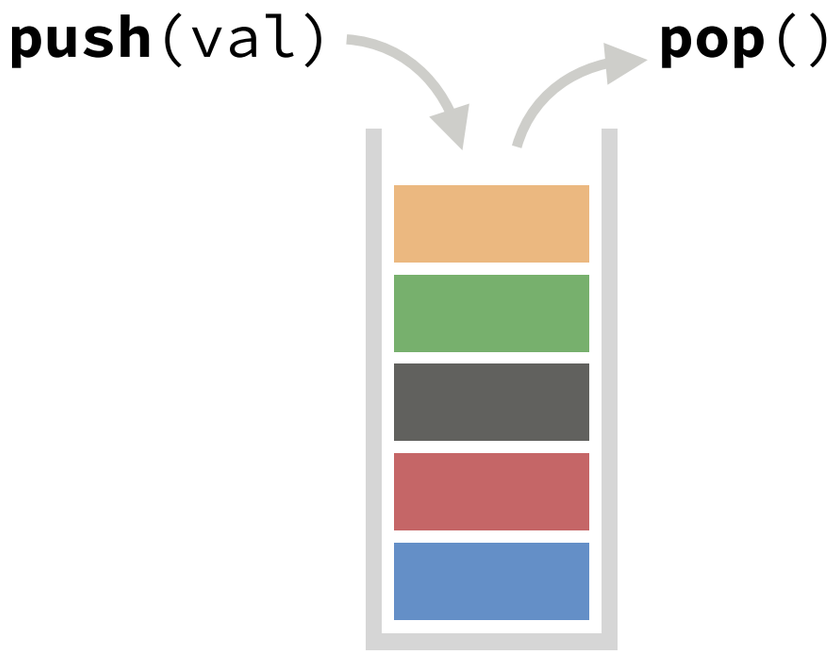
! Elements cannot be accessed or modified in any other way



LIFO: Last In First Out

~~**FILO**: First In Last Out~~

Stacks (a new ADT)

Stack

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void push(const T& val)
    T pop()
    T top() const
```

add `val` to the collection of elements

remove the most recently added element

return the most recently added element

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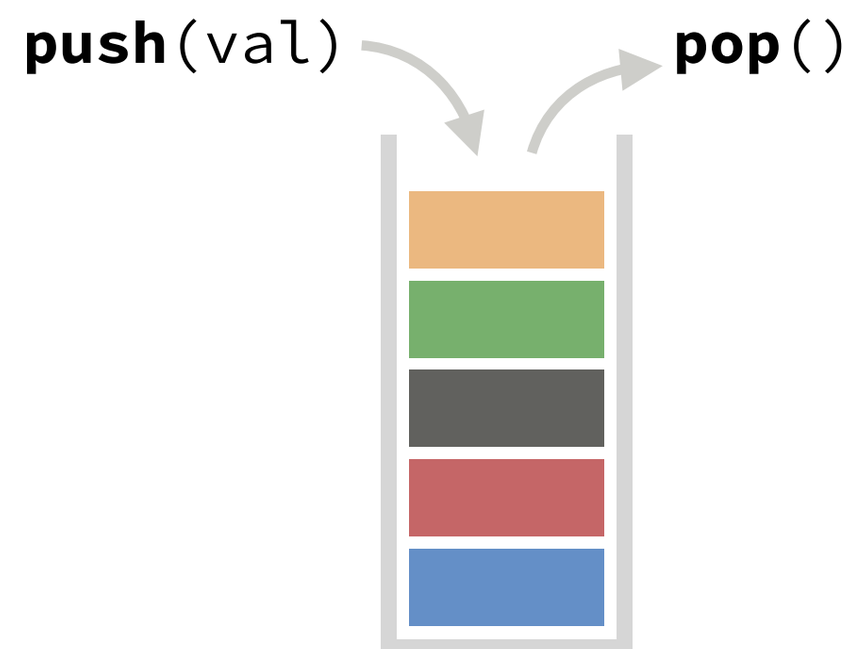
مكـــدوس

Eat this first

Inaccessible until the makdous above is eaten.



LIFO: Last In First Out

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Exercise

What does the following code do?

```
void easy_to_trace(int a[], int size)

    Stack<int> s;
    for (int i = 0; i < size; i++)
        s.push(a[i]);

    for (int i = 0; i < size; i++)
        cout << s.pop() << endl;
```

Exercise

What does the following code do?

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    Stack<int> s;
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        cout << s.pop() << endl;
```

Answer. Prints the array elements in reverse order.

Application # 1

Stacks are used to ***reverse*** the order of a sequence.

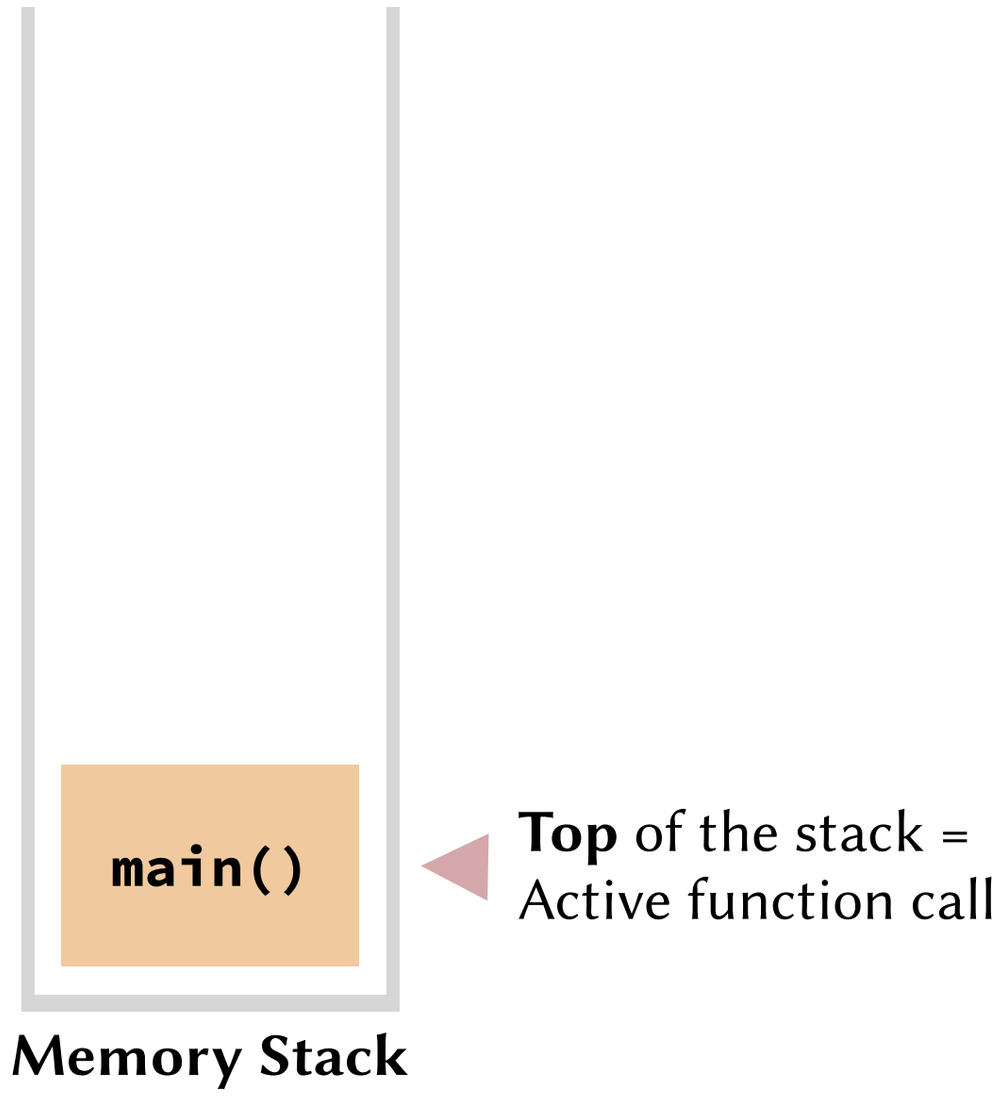
Stacks (Applications)

1. Reversing sequences.
2. **Managing function calls.**

```
void f2() {
    ...
}

void f1() {
    ...
    f2();
    ...
}

int main() {
    ...
    f1();
    ...
}
```



Stacks (Applications)

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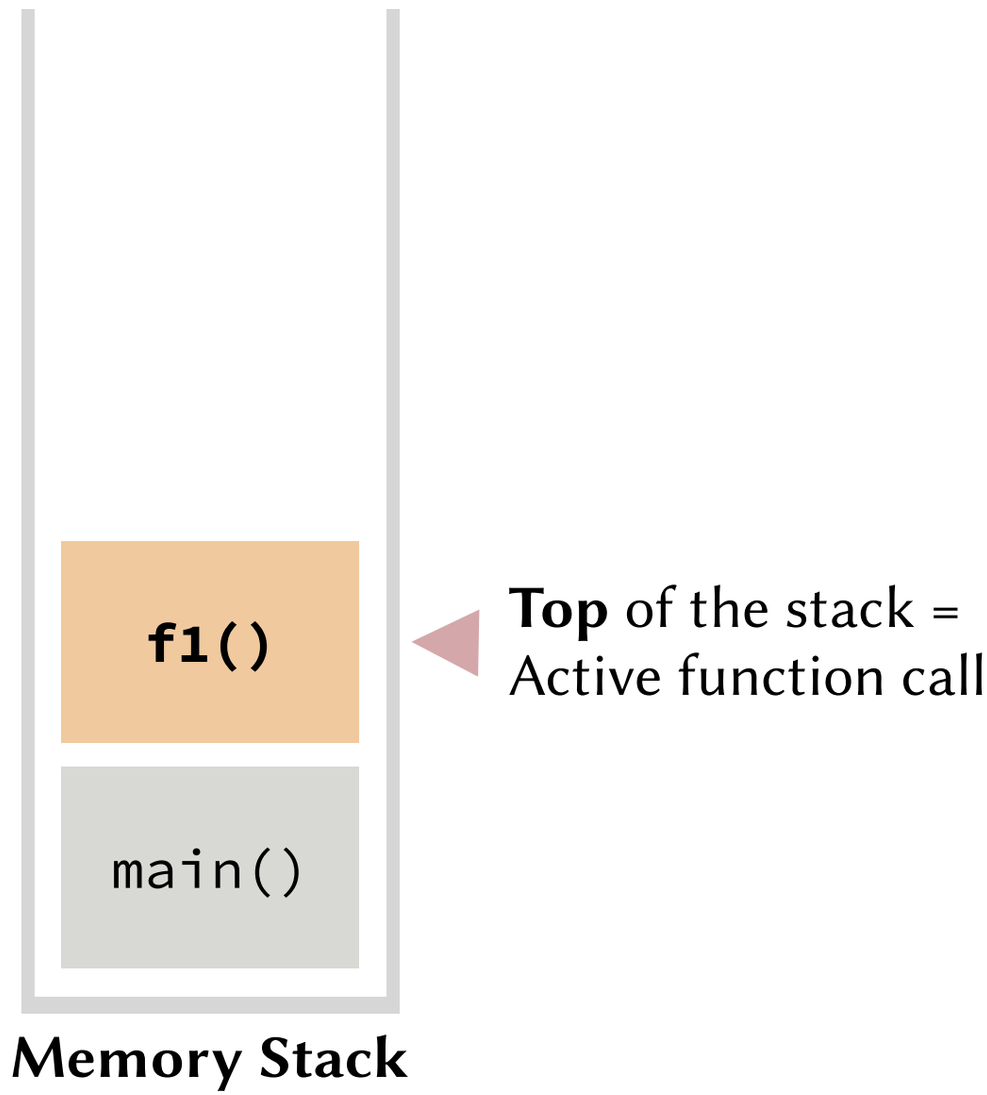

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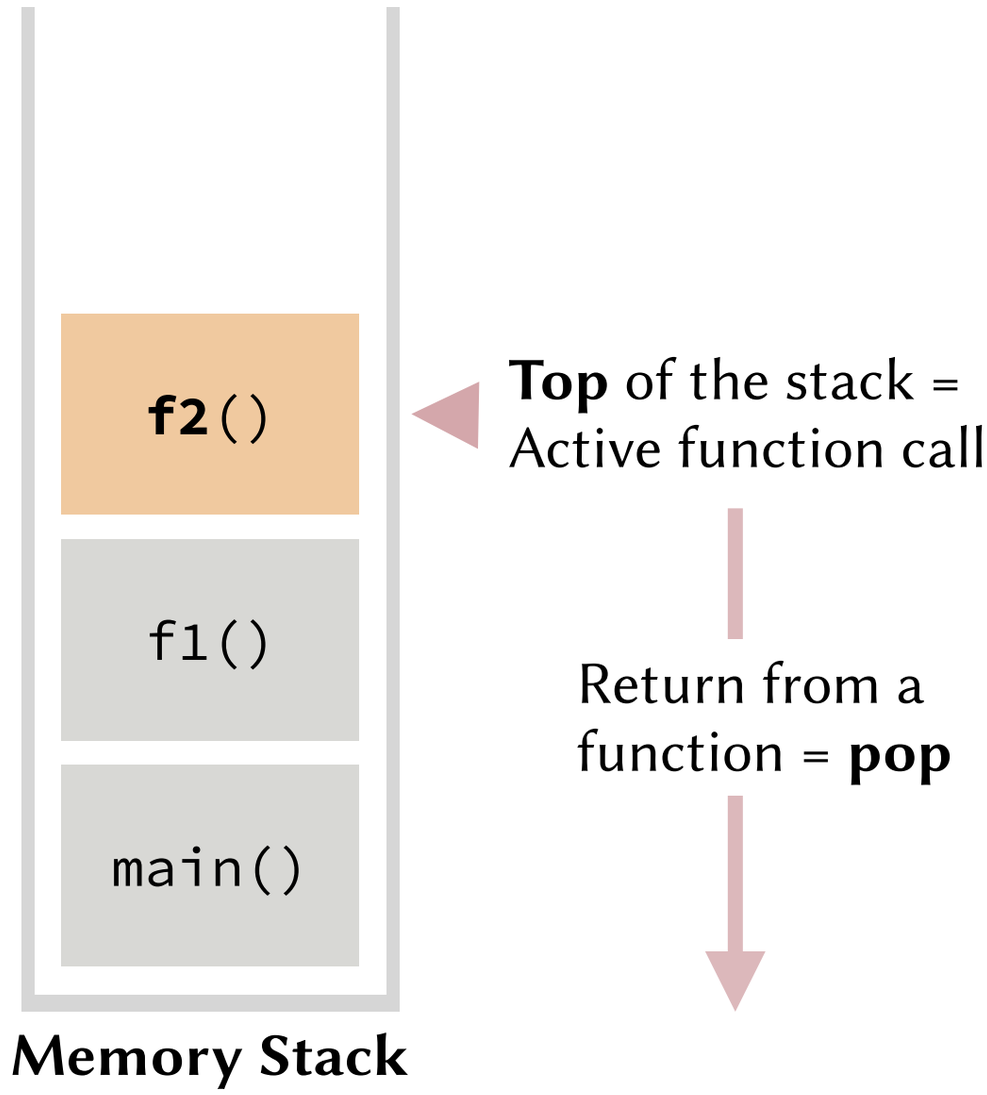

Stacks (Applications)

1. Reversing sequences.
2. **Managing function calls.**

Since **recursion** uses the memory stack, recursive algorithms typically can be converted to iterative algorithms using an explicit stack.

Stay tuned for examples when we cover the binary search tree data structure.

```
void f2() {
    ...
}

void f1() {
    ...
    f2();
    ...
}

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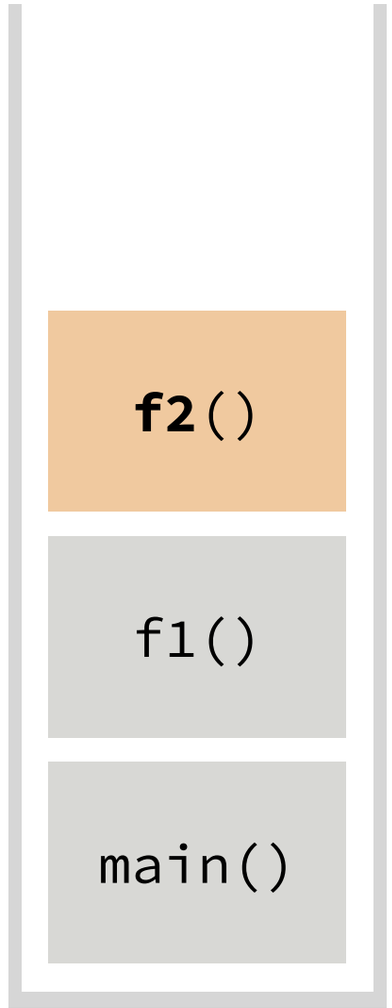

Memory Stack

Top of the stack = Active function call

Return from a function = **pop**

Stacks (Applications)

1. Reversing sequences.
2. Managing function calls.
3. **Undo and Redo in a word processor**
   Back and Forward in a web browser

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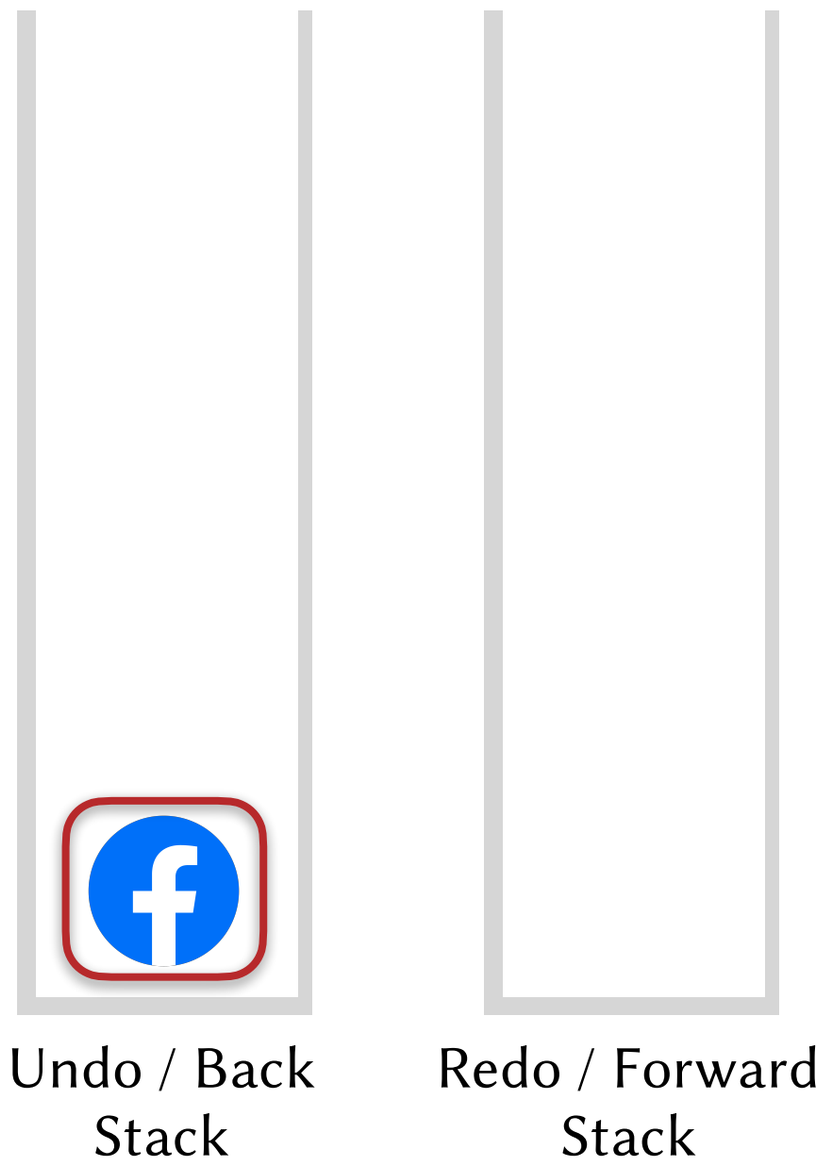


```
visit(page)

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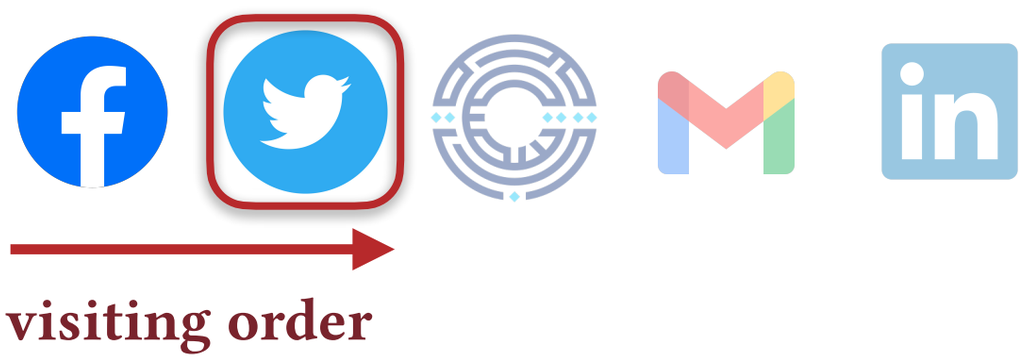

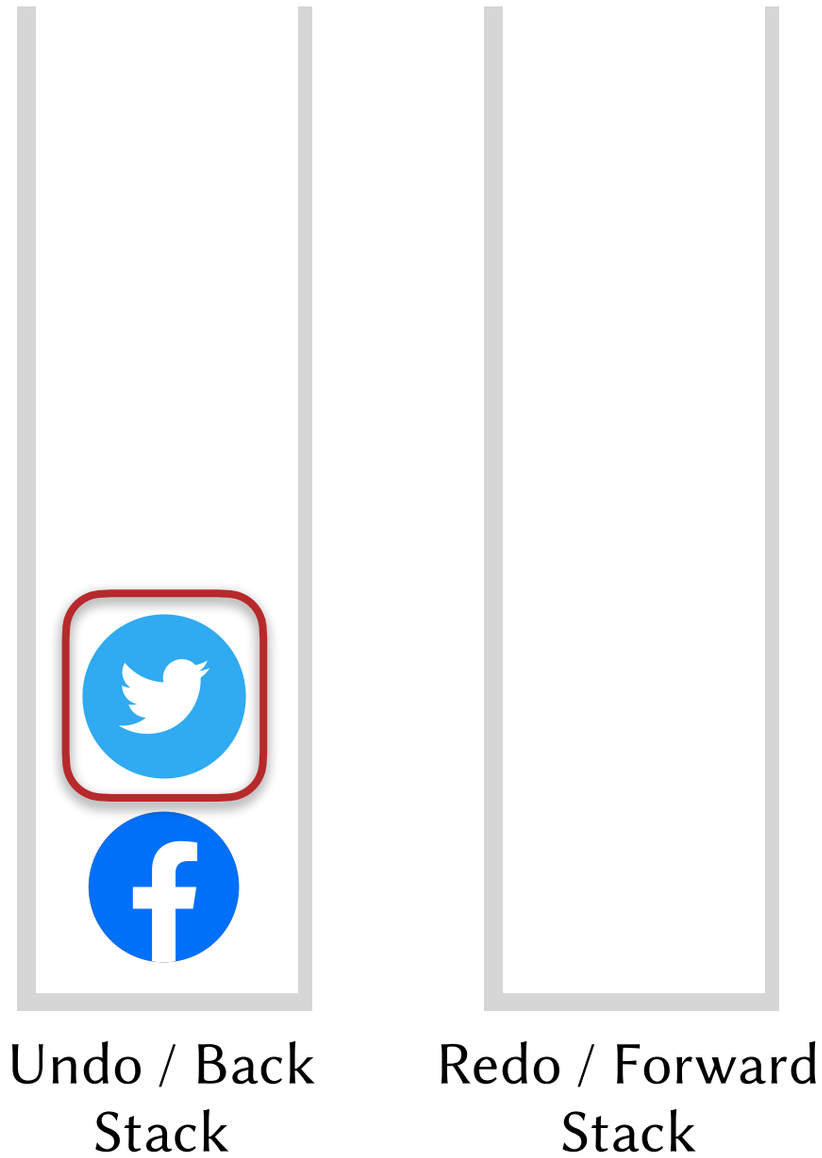


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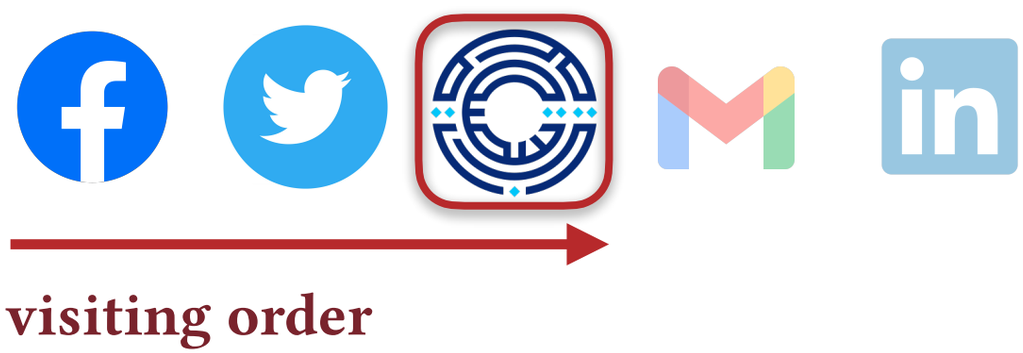

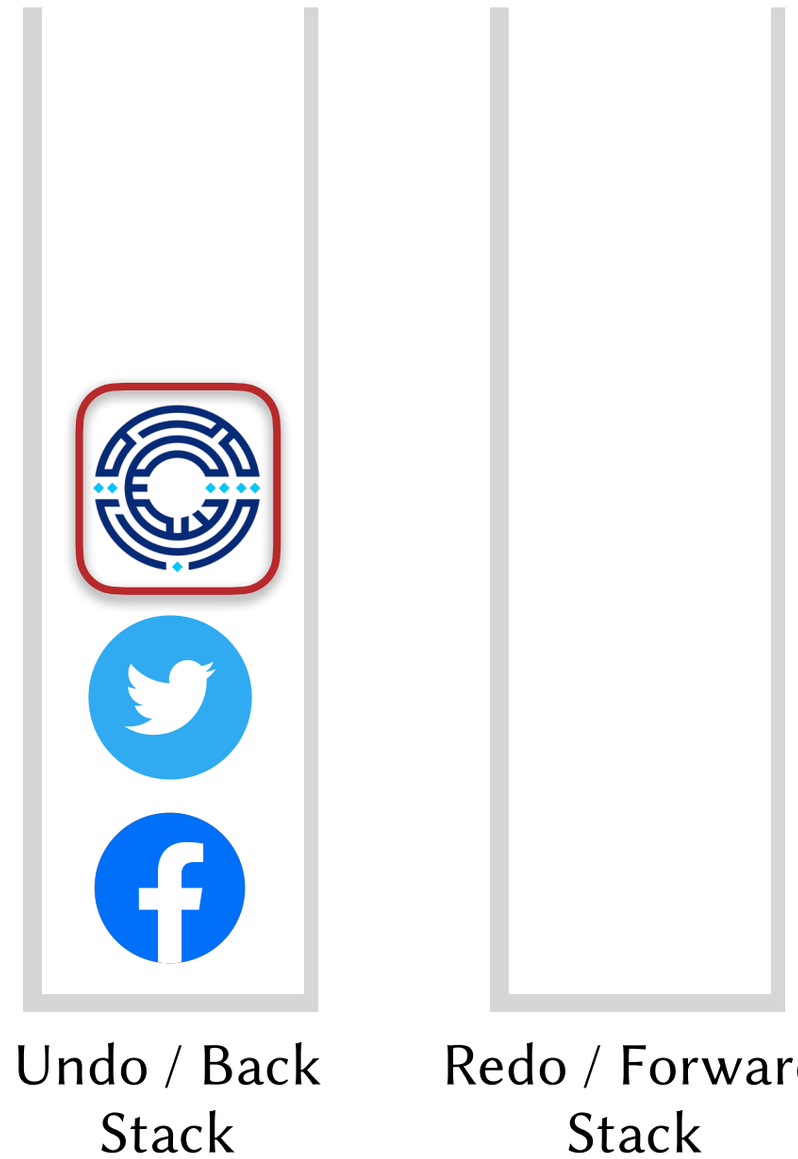


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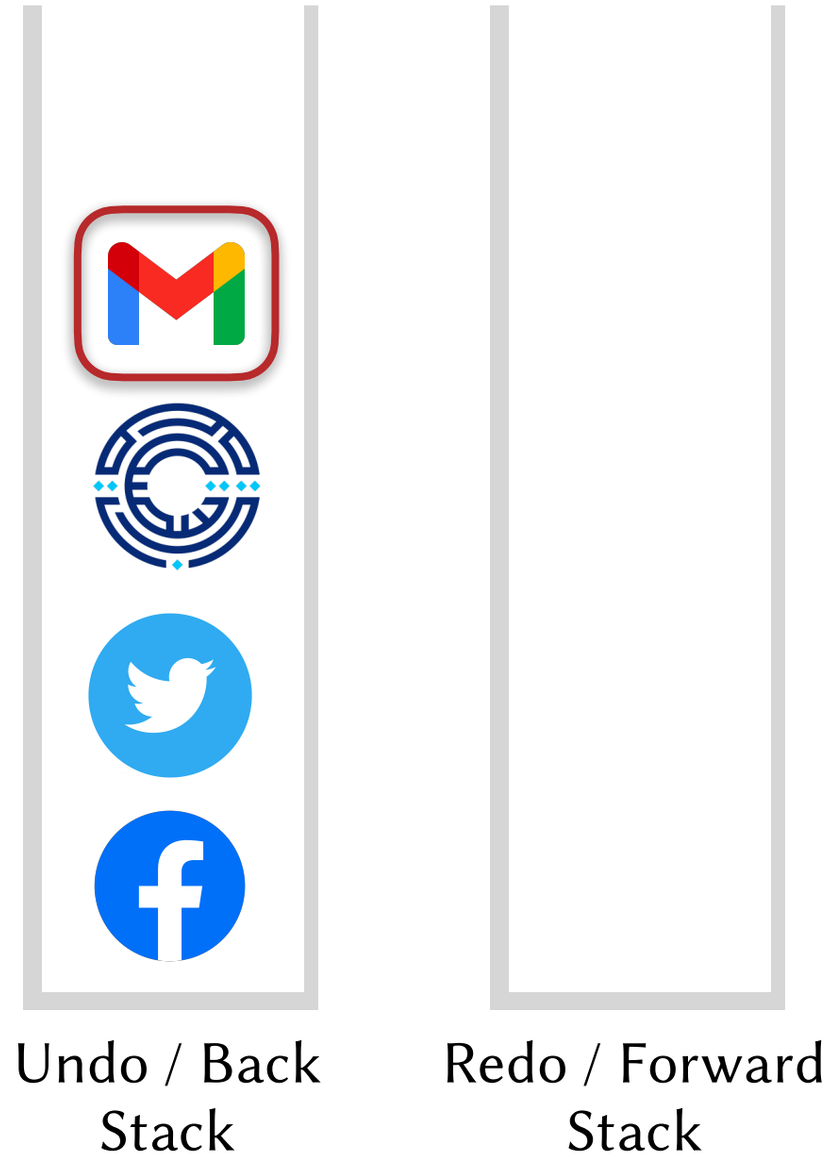


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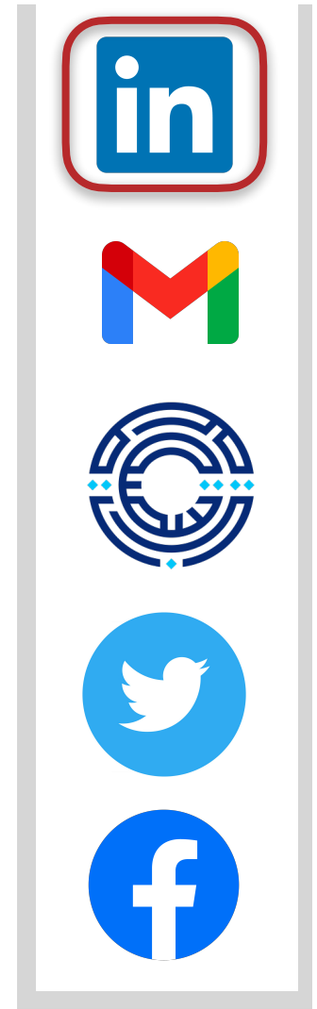


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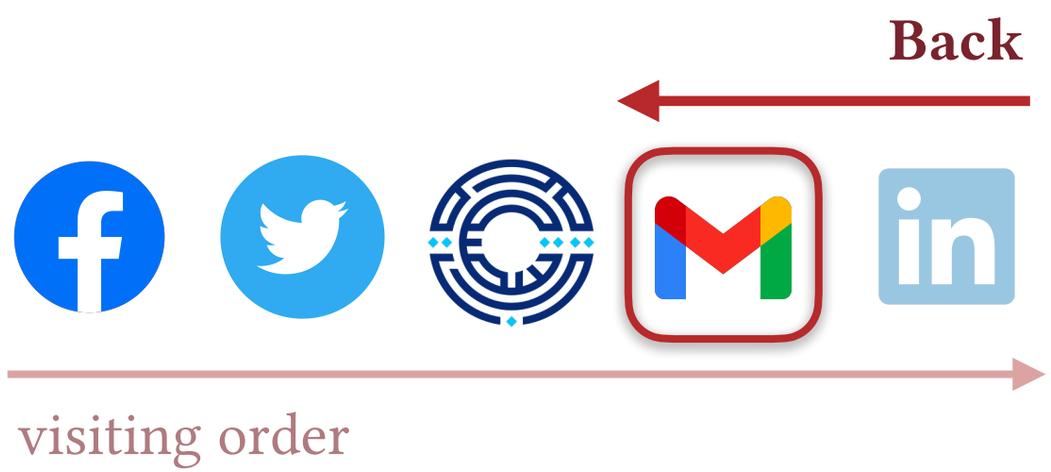

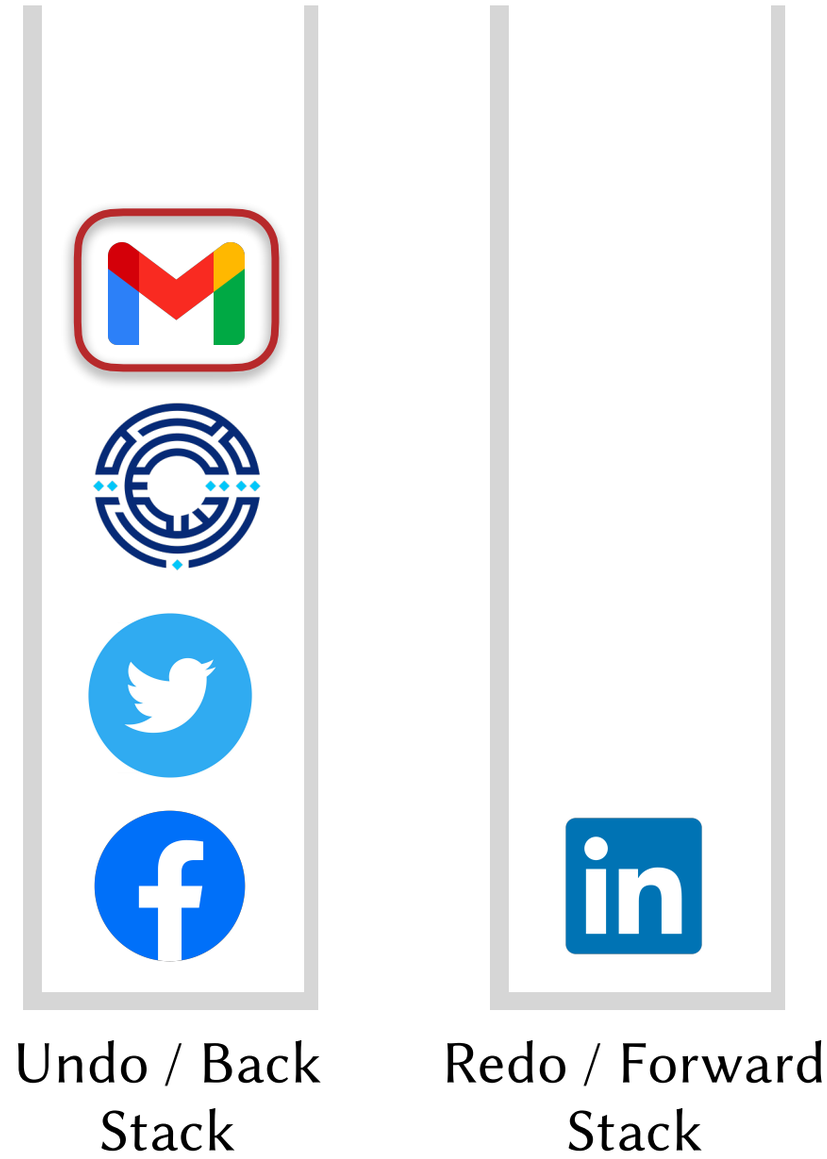


```
visit(page)
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```

```
back()
    redo.push(undo.pop())
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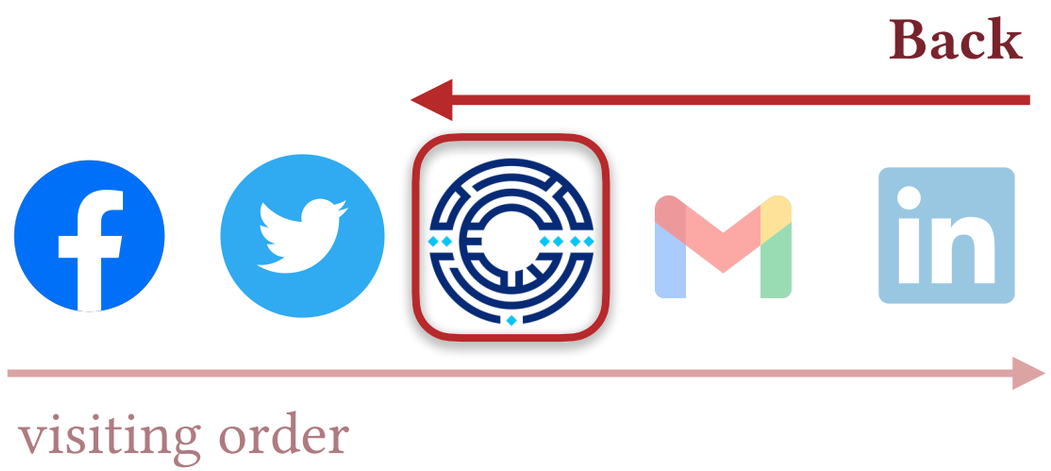

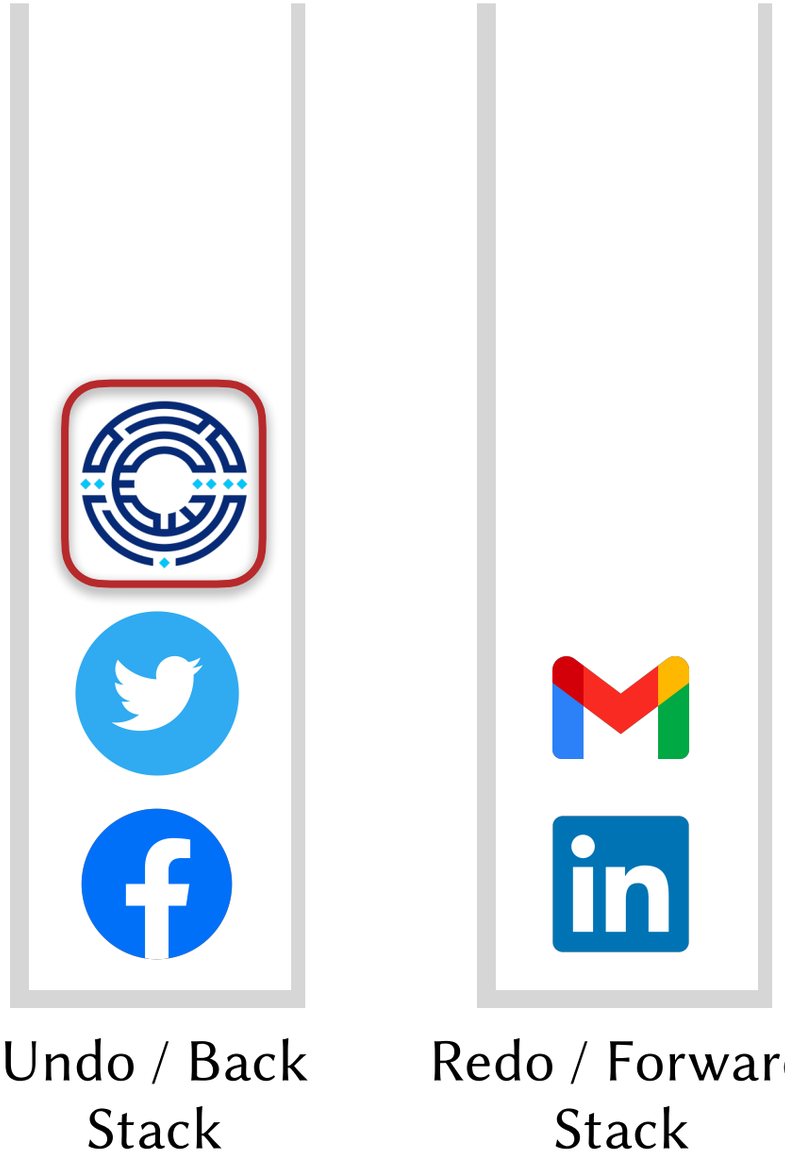


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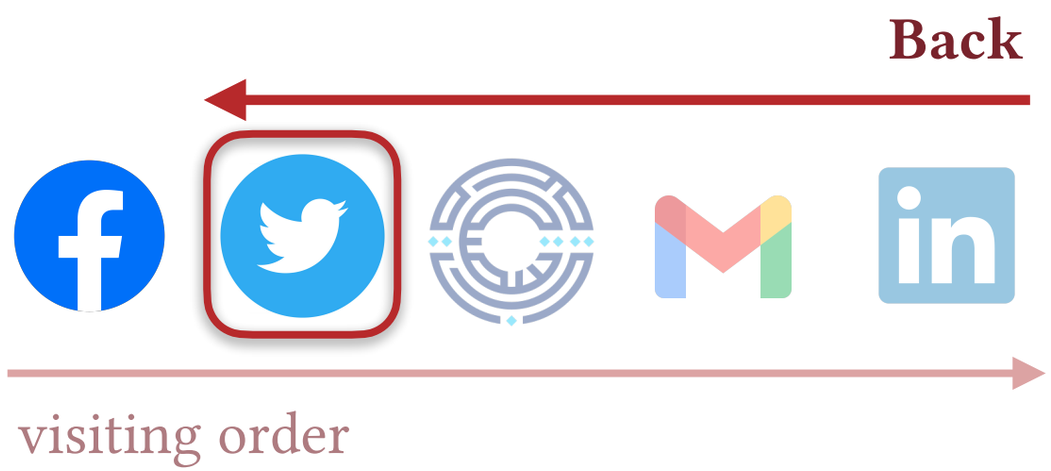

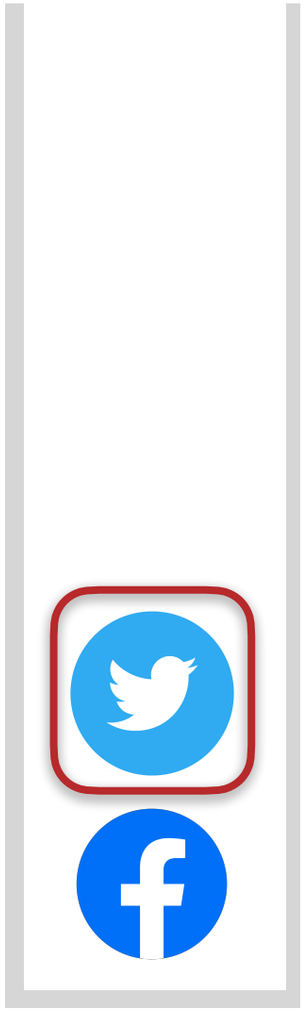

Undo / Back Stack

Redo / Forward Stack

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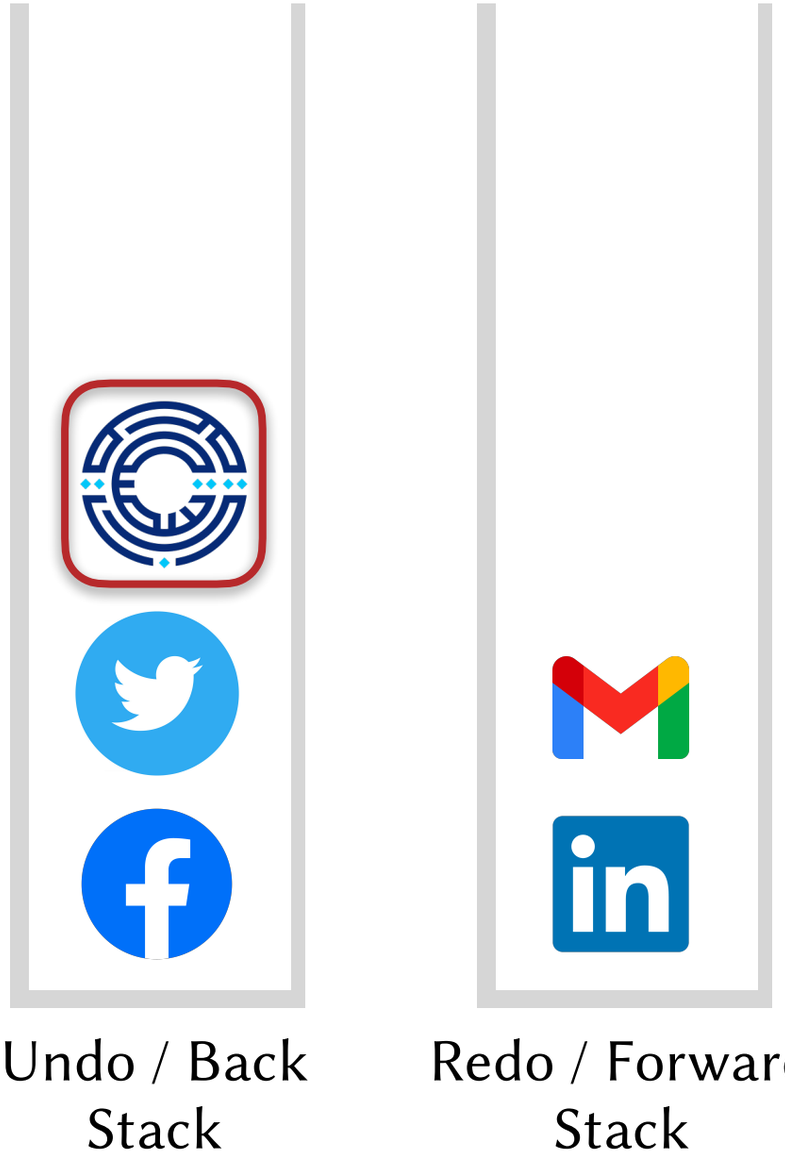


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Stacks (Applications)

1. Reversing sequences.
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   Back and Forward in a web browser
4. **Balancing parentheses.**

   - `()[{}]([]{})` **VALID**
   - `((([{}]{[]{}}` **INVALID** (opened but not closed)
   - `[{}]))){[]{}}` **INVALID** (closed but not opened)
   - `[(]){()[}][]` **INVALID** (misbalanced)

Stacks (Applications)

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   Back and Forward in a web browser
4. **Balancing parentheses.**

- `()[{}]([]{})` **VALID**
- `(([{}]{[]{}}` **INVALID**
- `[{}]))){[]{}}` **INVALID**
- `[(]){()[}][]` **INVALID**

CHECK_PARANTHS(string str)

```
for (int i=0; i<str.length(); i++) {
    if ( /* str[i] is (, {, or [ */ )
        stack.push(str[i]);
    else if (stack.is_empty() ||
            /* str[i] does not
             match stack.top() */ )
        return false;
    else
        stack.pop();
}
return stack.is_empty();
```

Try **tracing** the code!

Exercise

Which of the following are reasonable implementations for the Stack ADT?

- **Doubly-linked** list:
```
push(val) = add_to_head(val)
     pop() = remove_head()
```
- **Doubly-linked** list:
```
push(val) = add_to_tail(val)
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```
- **Singly-linked** list:
```
push(val) = add_to_tail(val)
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- **Singly-linked** list:
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```
- **Array** list:
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```
push(val) = add_to_tail(val)
     pop() = remove_tail()
```

pop() = $O(n)$

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pop() = $O(n)$
push() = $O(n)$

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😁 **Singly-linked** list:

```
push(val) = add_to_head(val)
     pop() = remove_head()
```

Uses less memory than DLList & avoids array resizing!

✗ **Array** list:

```
push(val) = add_to_head(val)
     pop() = remove_head()
```

pop() = $O(n)$
push() = $O(n)$

😁 **Array** list:

```
push(val) = add_to_tail(val)
     pop() = remove_tail()
```

Makes good use of cache memory and avoids creating and deleting nodes.

Stack as a DLL (Implementation)

```
template <class T>
class Stack {
private:
    DLList<T> list;

public:
    bool is_empty() const     { return list.is_empty(); }
    void clear()              { list.clear(); }
    void push(const T& val)   { list.add_to_head(val); }
    T top() const             { return list.head_val(); }

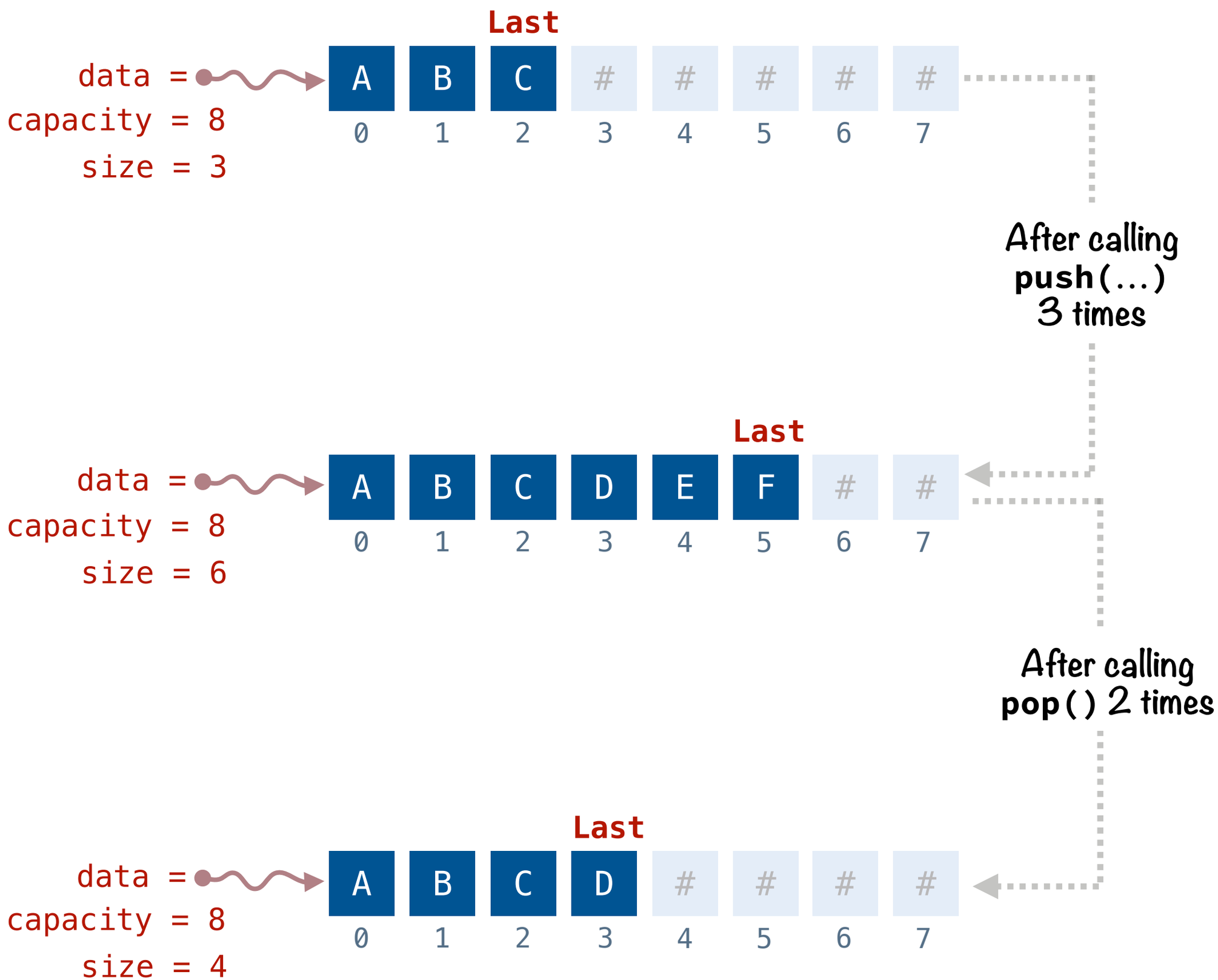
    T pop() {
        T val temp = list.head_val();
        list.remove_head();
        return temp;
    }
};
```

delegate all the work to this object

throws an exception if the list is empty!

No need to implement a default constructor, a copy constructor, or a destructor, or overload the assignment operator because this was already done in class `DLList`!

Stack as an **Array** (Implementation)



Exercise

Which of the following are reasonable implementation for function **`clear()`**?

A.
```
void clear() { last = -1; }
```

B.
```
void clear() {
    last = -1;
    delete [] data;
}
```

C.
```
void clear() {
    last = -1;
    delete [] data;
    data = new data[capacity];
}
```

D. None of the above.

Exercise

Which of the following are reasonable implementation for function **`clear()`**?

✓
```
void clear() { last = -1; }
```

The next `push(...)` statement moves last to index `0`

✗
```
void clear() {
    last = -1;
    delete [] data;
}
```

The array is not usable after this statement.

✗
```
void clear() {
    last = -1;
    delete [] data;
    data = new data[capacity];
}
```

Unnecessarily removes the array and crates a new one with the same capacity.

D. None of the above.

Queues (a new ADT)

Queue

```
void enqueue(const T& val)
   T dequeue()
   T get_first() const
   T get_last()  const
```

add `val` to the collection of elements

remove the first added element

return the first added element

return the last added element

! Elements cannot be accessed or modified in any other way

First come first serve

FIFO: Fast In First Out

~~**LILO:** Last In Last Out~~

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Applications.

1. Ticketing systems.
2. Printer queue.
3. Website requests queue.
4. Video streaming buffer.

etc. (too many to be listed)

FIFO: Fast In First Out

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Exercise

Which of the following are reasonable implementations for the Queue ADT?

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A Queue As An Array

Problem.

- A Queue must modify the head (either to remove or to add).
- Adding to or removing from the head of a regular array list requires shifting (costs $O(n)$)

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Solution.

- Avoid shifting: Store the index of where the elements start in the queue and where they end.
- Wrap-around if needed.

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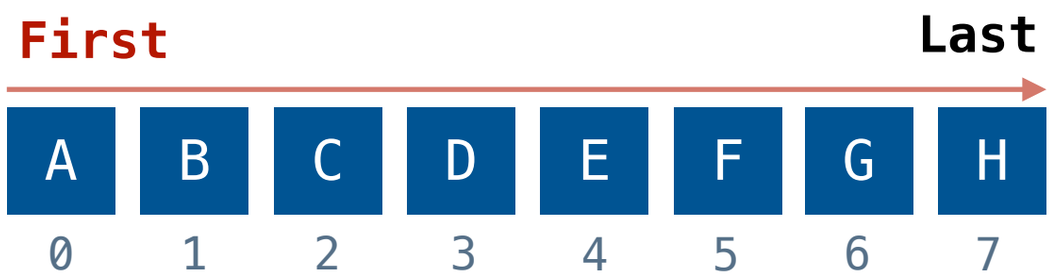
- Wrap-around if needed.

A full queue



A Queue As An Array

Problem.

- A Queue must modify the head (either to remove or to add).
- Adding to or removing from the head of a regular array list requires shifting (costs $O(n)$)

Solution.

- Avoid shifting: Store the index of where the elements start in the queue and where they end.
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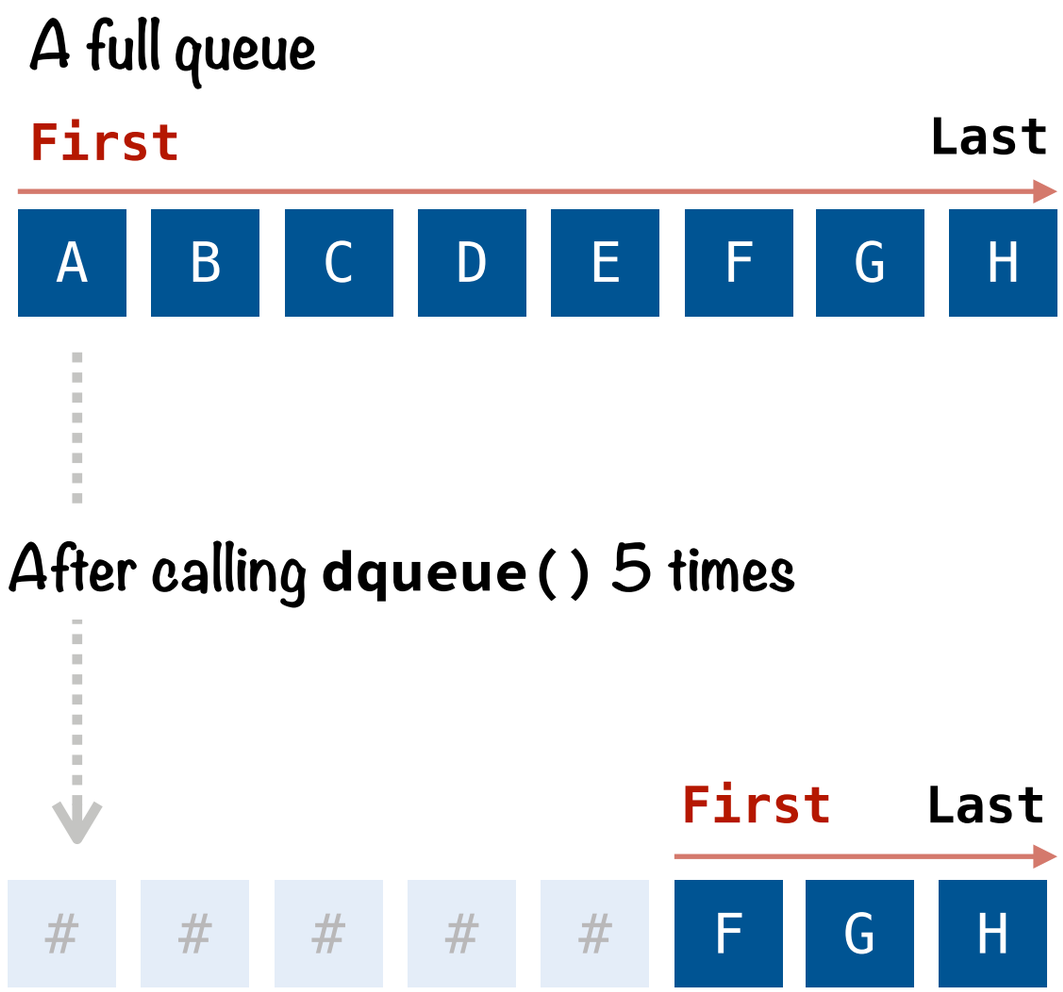

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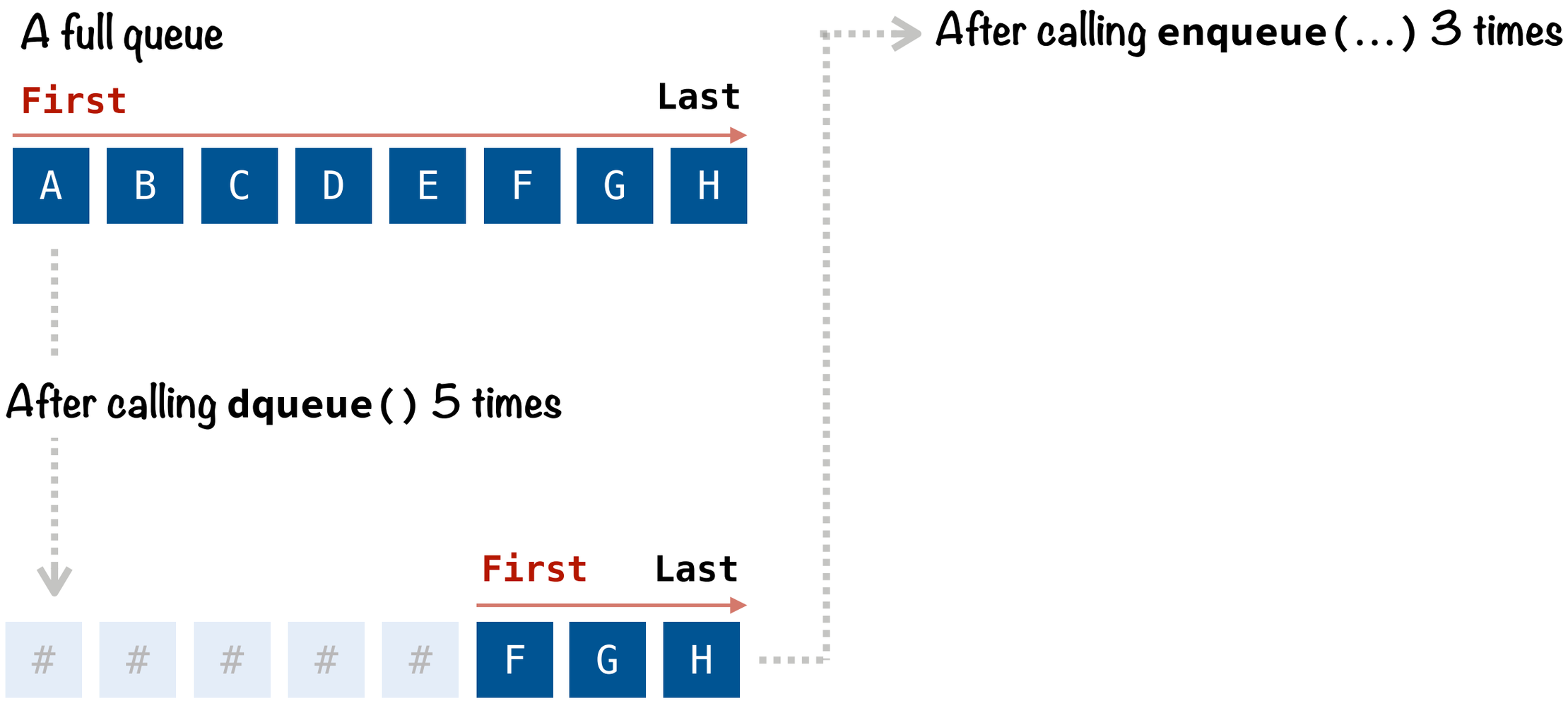

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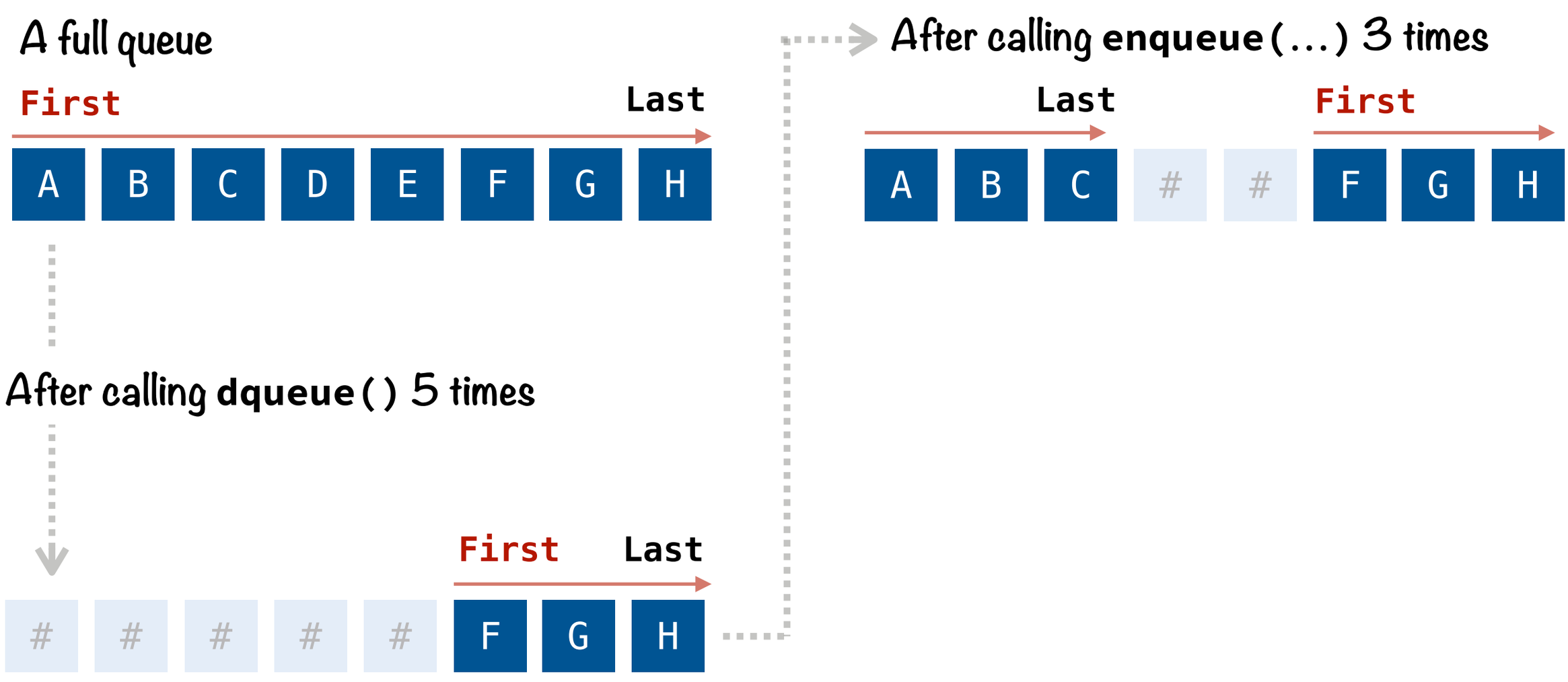

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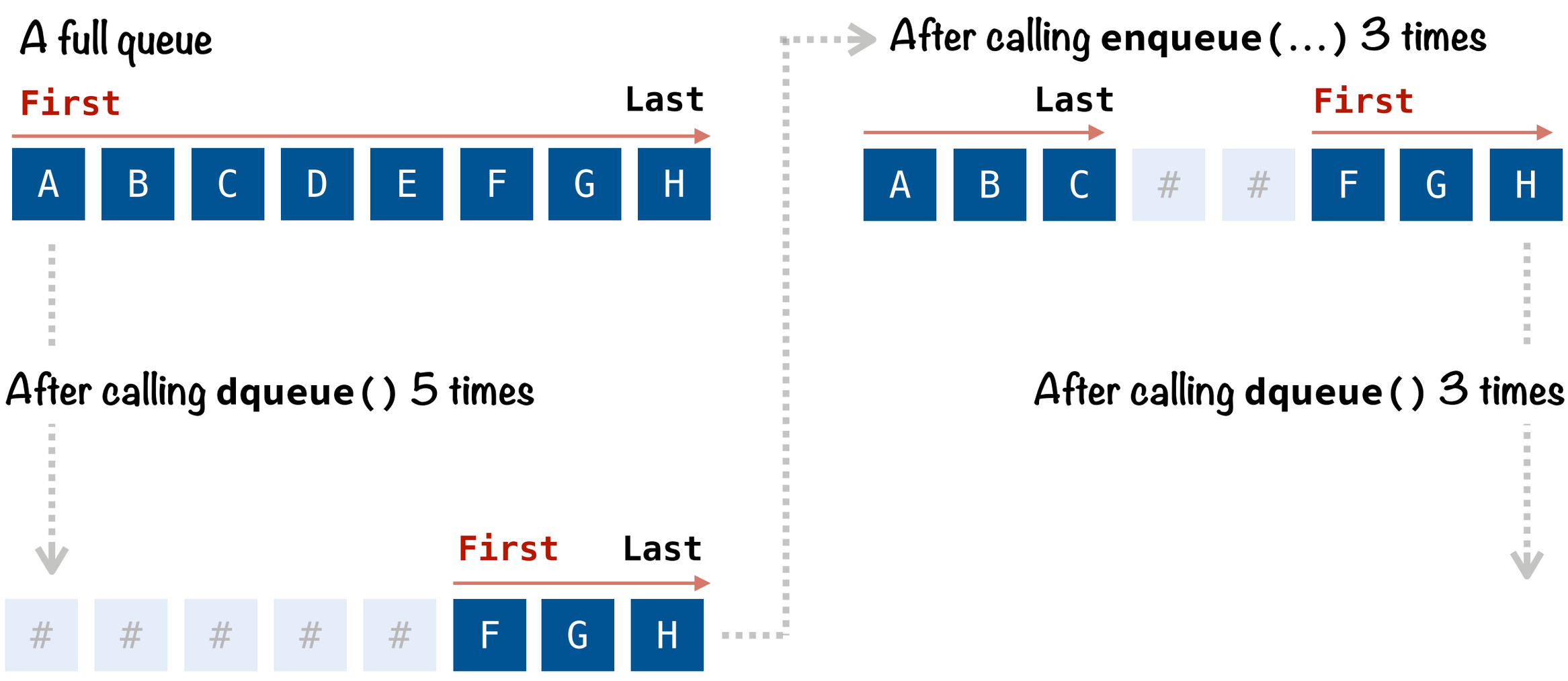

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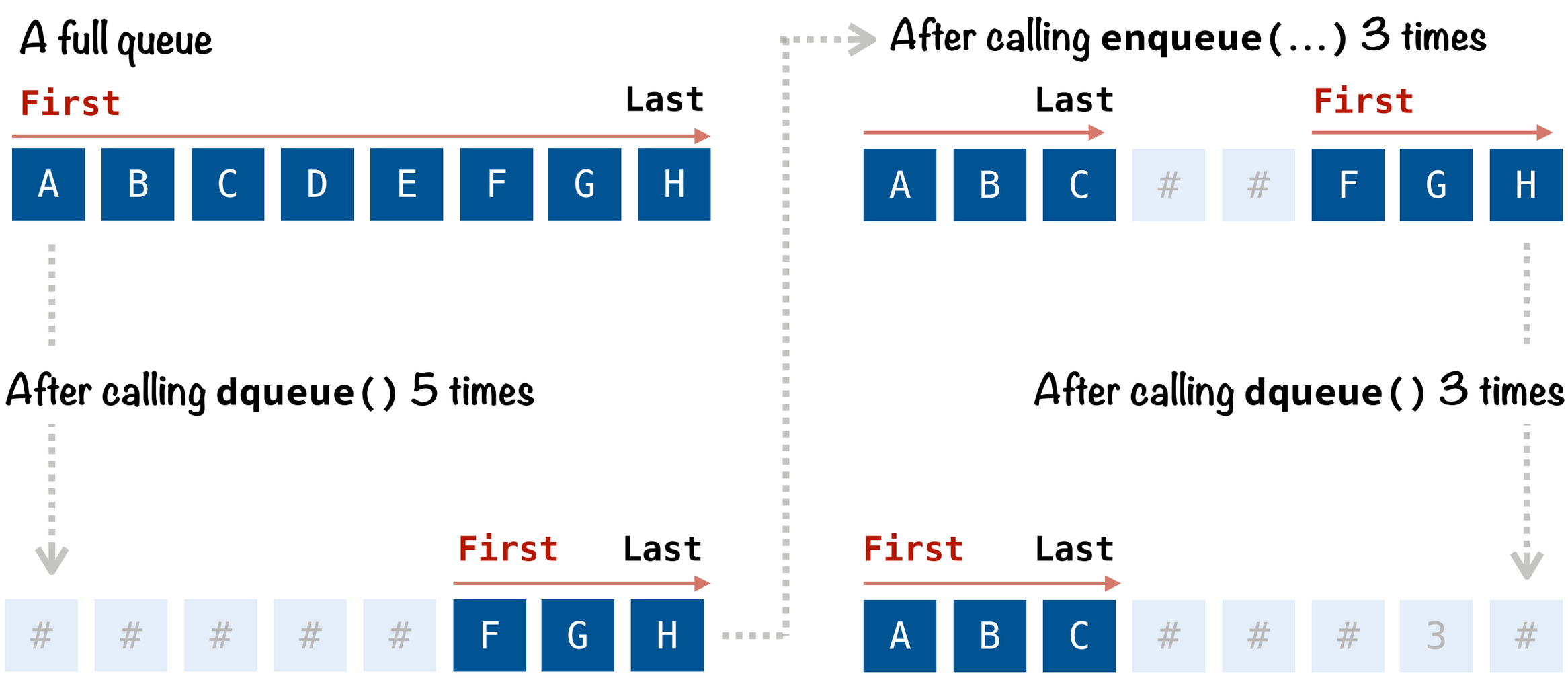
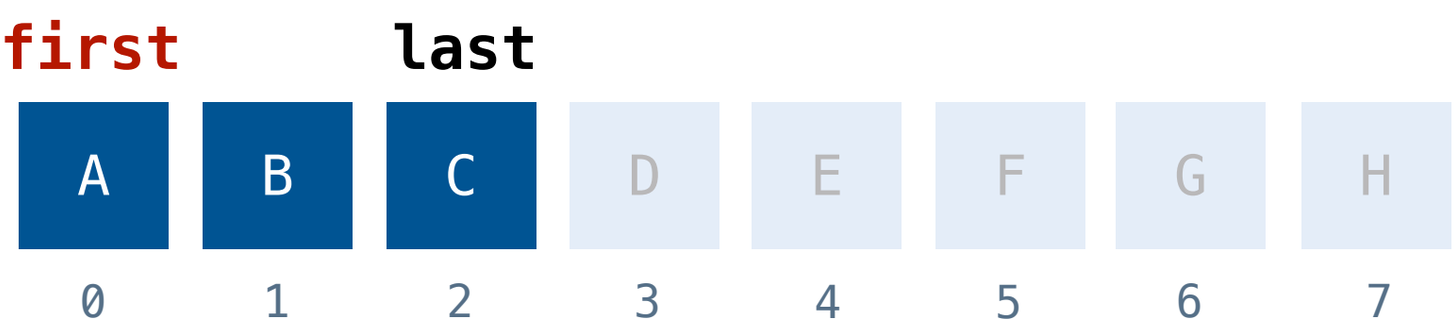


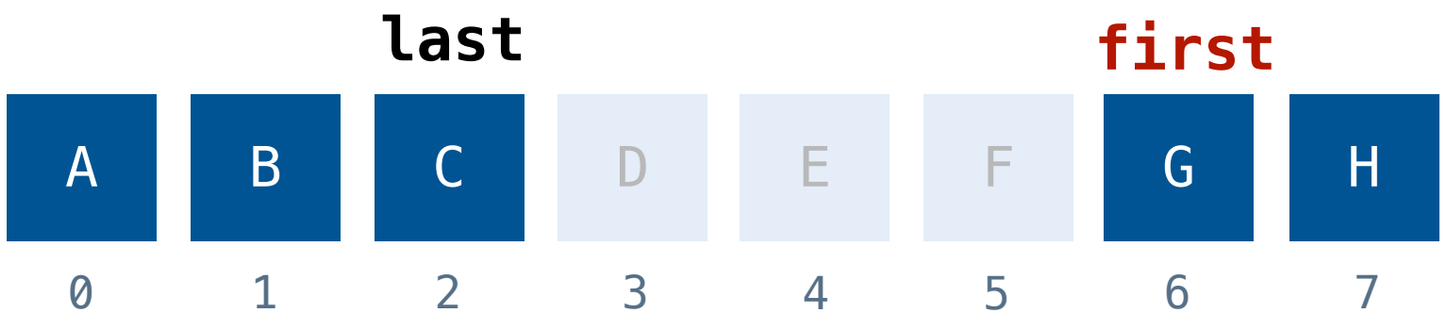
Queues as Arrays (data members)

first = -1
last = -1
capacity = 8
size = 0

capacity = 8
size = 3

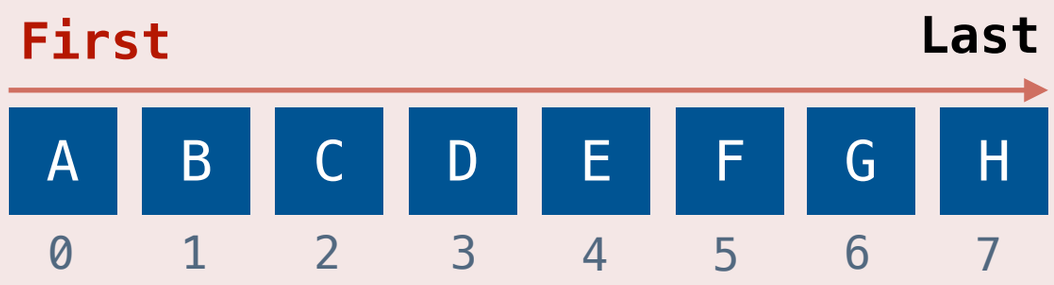


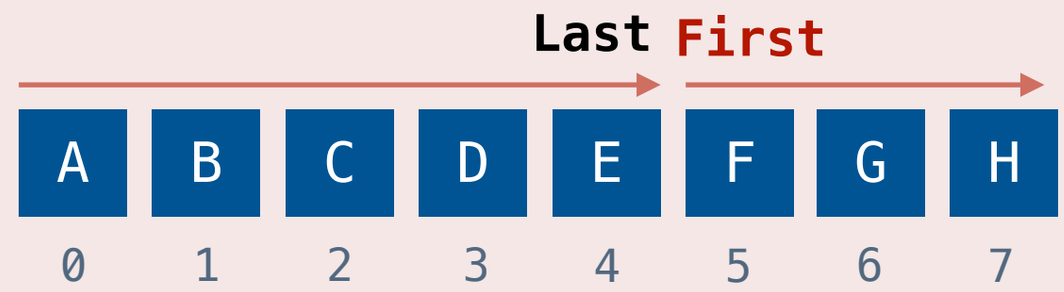
capacity = 8
size = 5

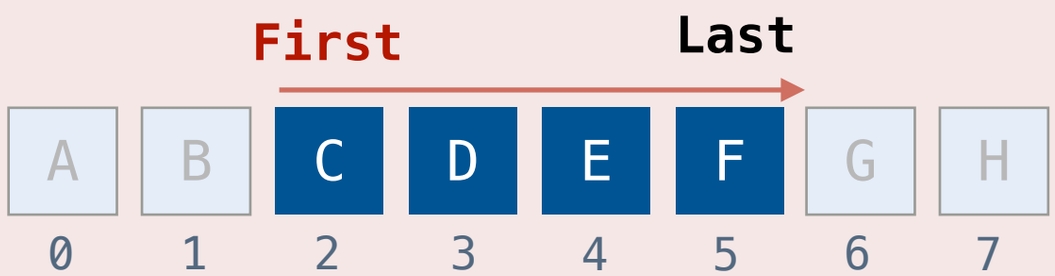


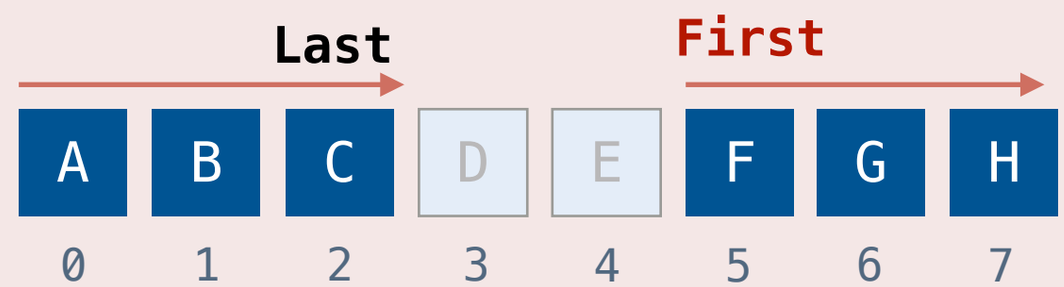
Exercise

Implement function **`is_full()`**









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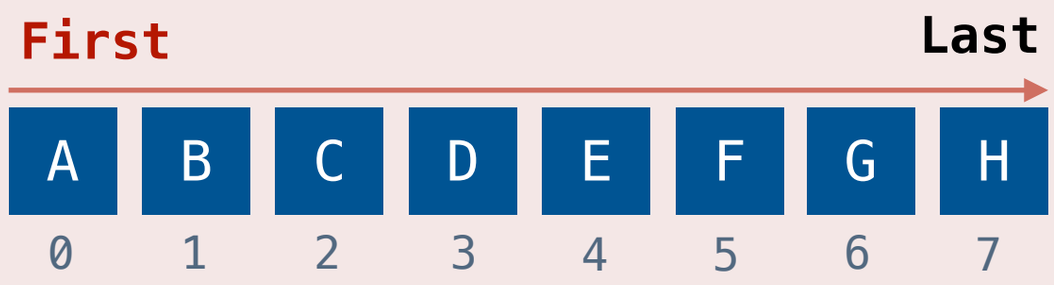
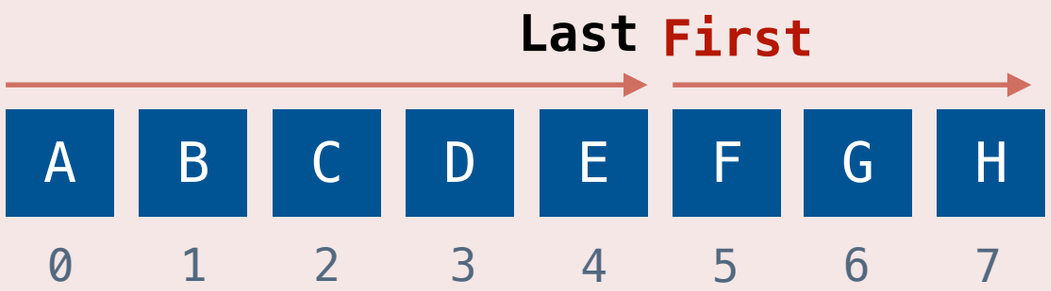
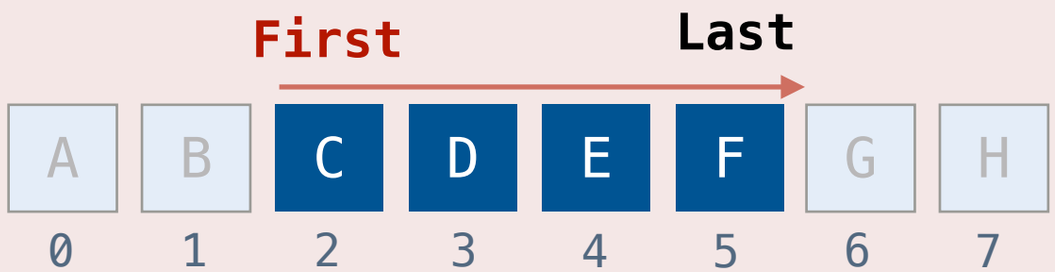
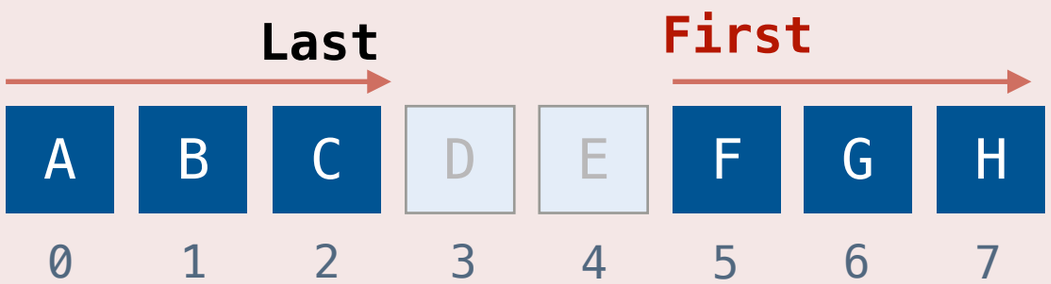








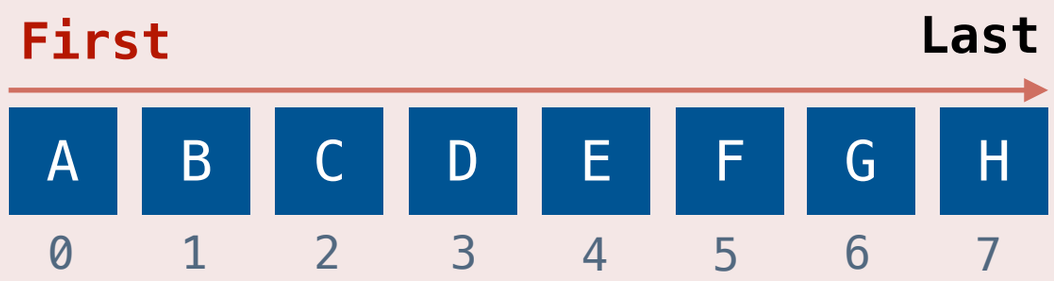

```
bool is_full() const {
    return (last+1) % capacity == first;
}
```

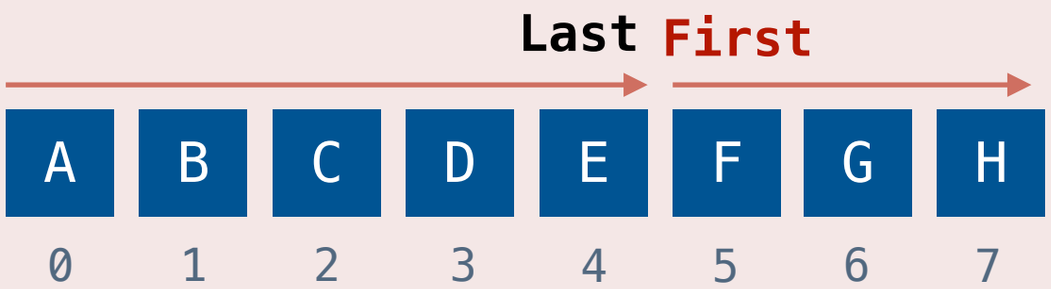
```
bool is_full() const {
    return size == cap;
}
```

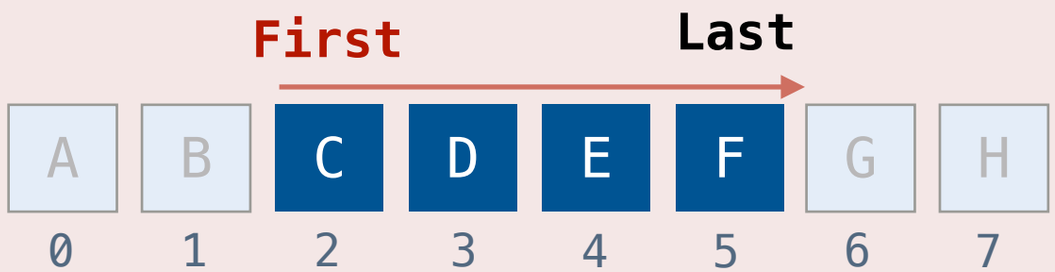
```
bool is_full() const {
    return
    last-first+1 == capacity ||
          last+1 == first;
}
```

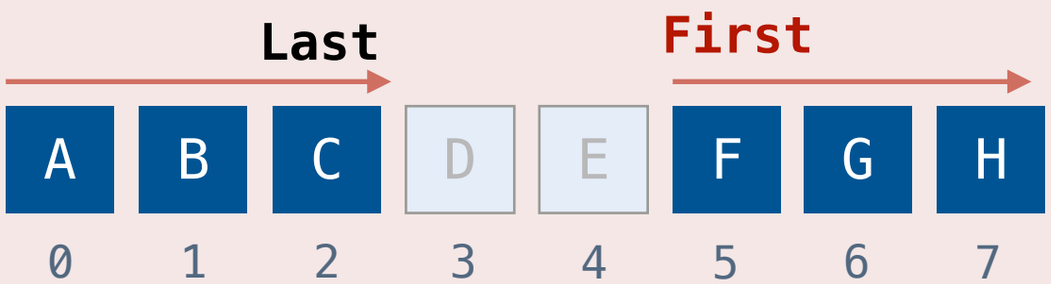
Exercise

Implement function **`is_full()`**







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    return size == cap;
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```

```
bool is_full() const {
   return
   last-first+1 == capacity ||
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}
```

Buggy code!

What if the queue is empty and capacity = 1?

Queues (Implementation Plan)

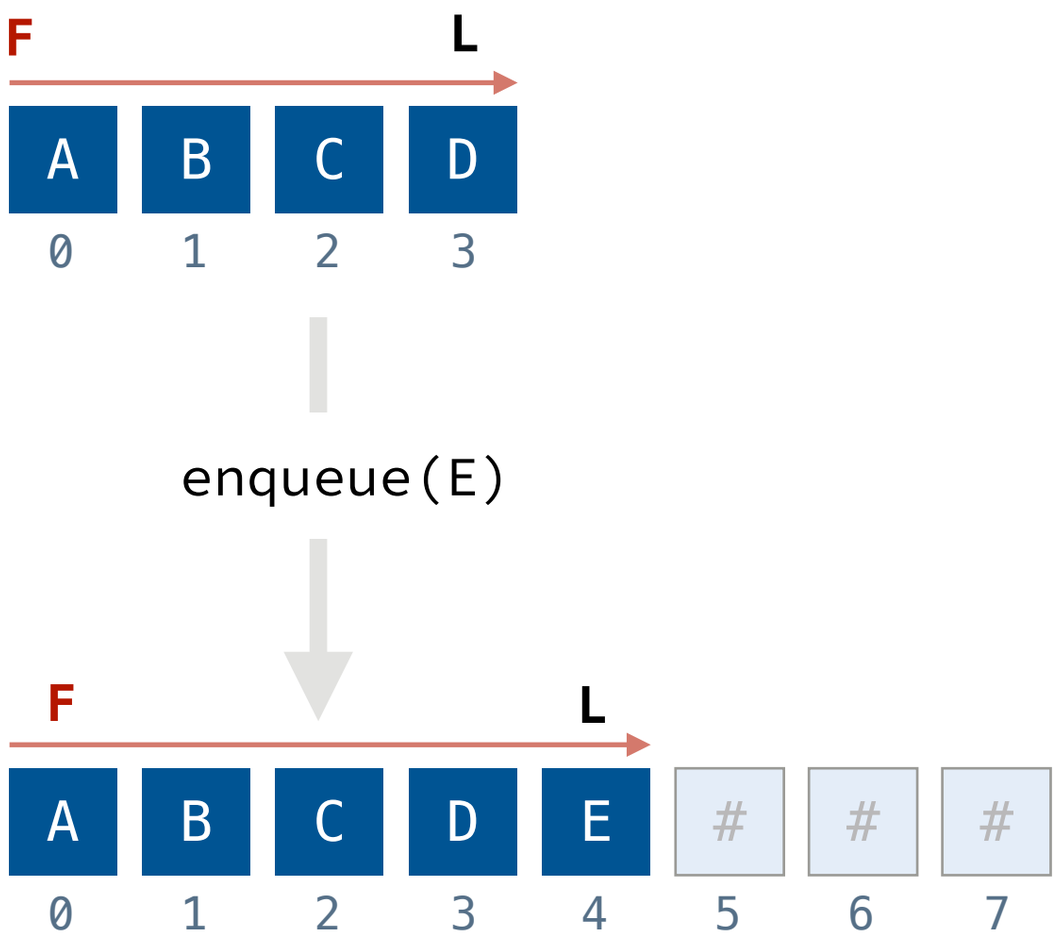
Resize up if full

```
last = (last + 1) % capacity;
data[last] = val;
```

Queue

```
void enqueue(const T& val)

    T dequeue()
```



Queues (Implementation Plan)

Queue

```
void enqueue(const T& val)

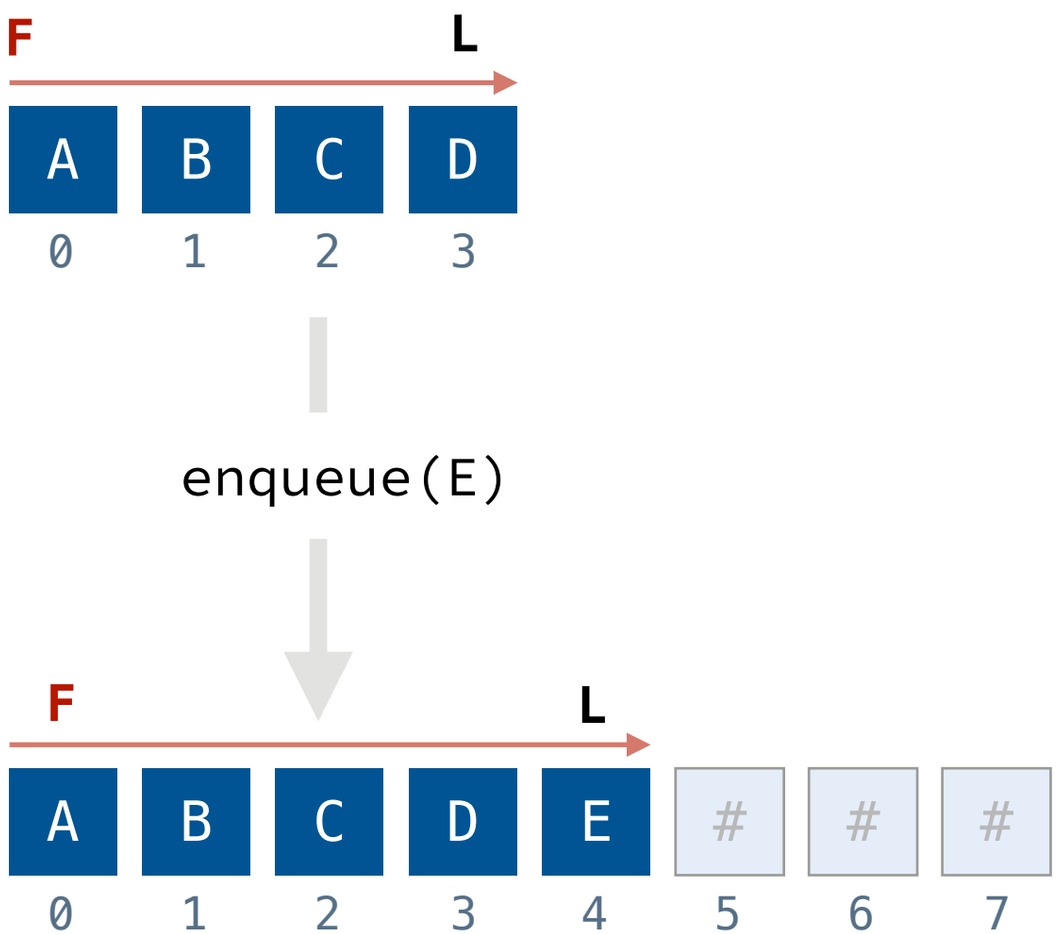
   T dequeue()
```

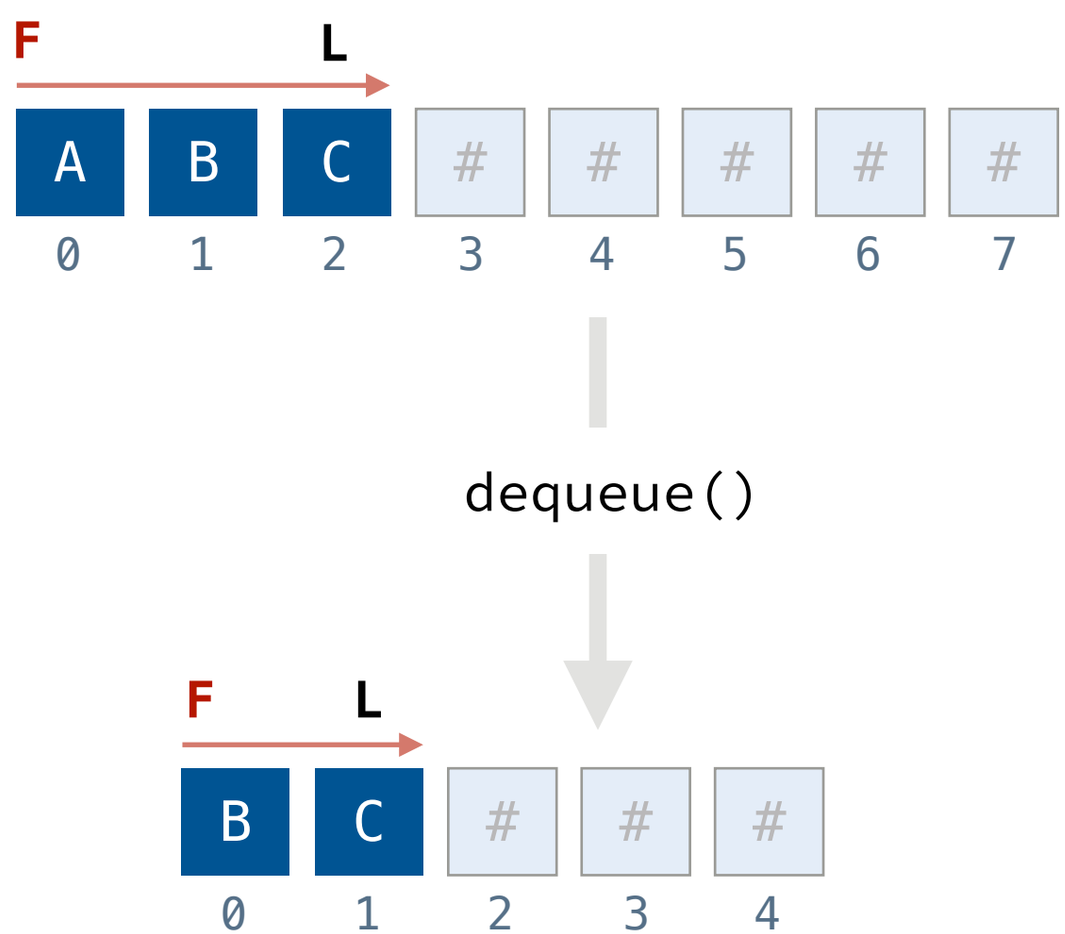
Resize up if full

```
last = (last + 1) % capacity;
data[last] = val;
```

```
first = (first + 1) % capacity;
```

Resize down if $\frac{1}{4}$ full





Exercise

What is the running time of the following code? Assume the queue is initially of size 1.

```
for (int i = 0; i < n; i++)
    myQueue.enqueue(i);
```

Choose the *best* answer.

A. $O(n^2)$

B. $O(n \log n)$

C. $O(n)$

D. $O(\log n)$

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A. $O(n^2)$

B. $O(n \log n)$

Too pessimistic!

$O(n)$

Correct!

D. $O(\log n)$

Incorrect

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What is the running time of the following code? Assume the queue is initially of size 1.

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x 1

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What is the running time of the following code? Assume the queue is initially of size 1.

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- [x] 1
- [x] [x] **1** (insert) + **1 (resize)**

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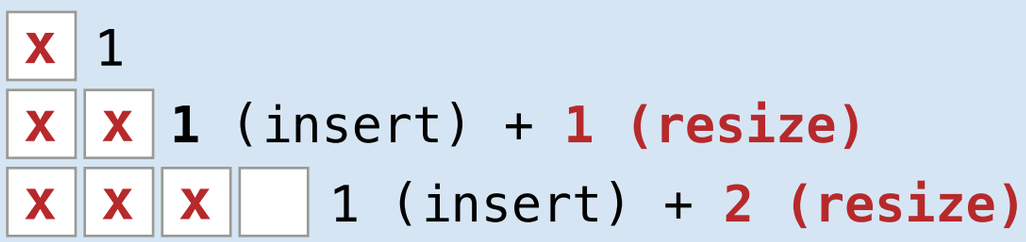

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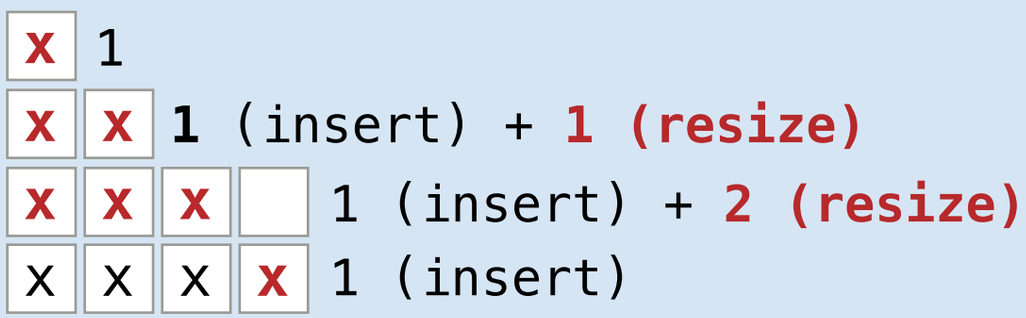

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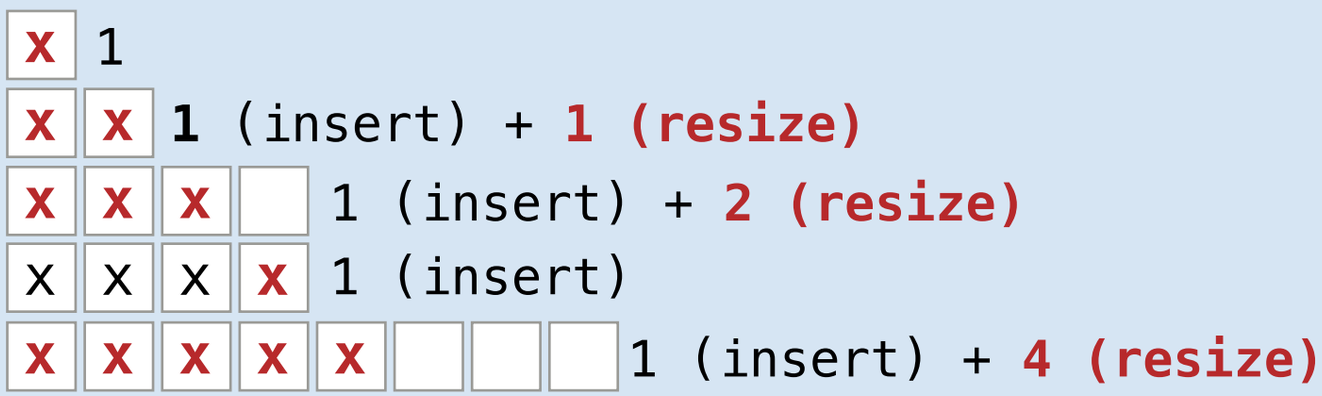

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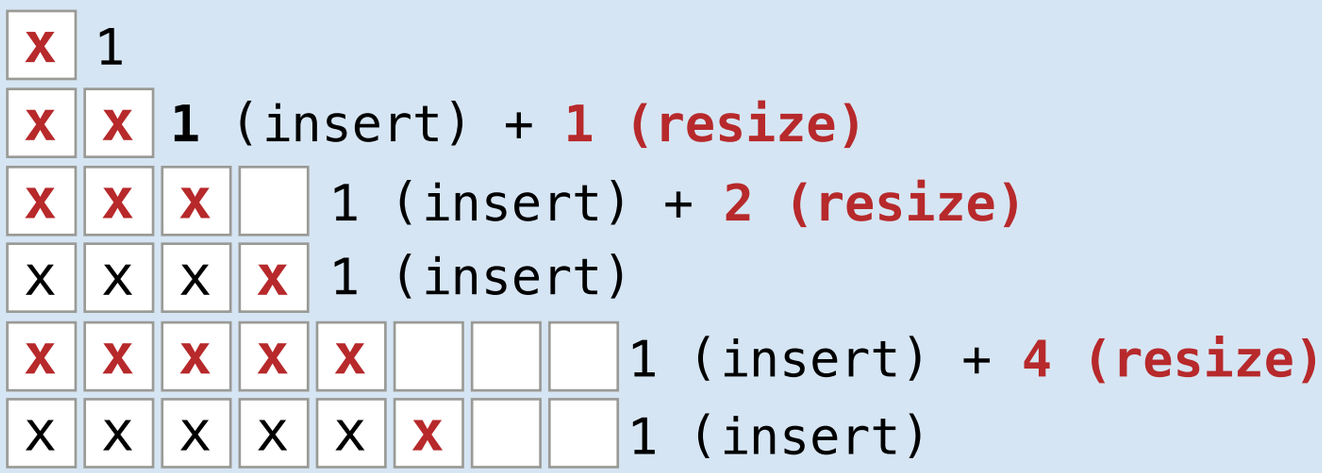

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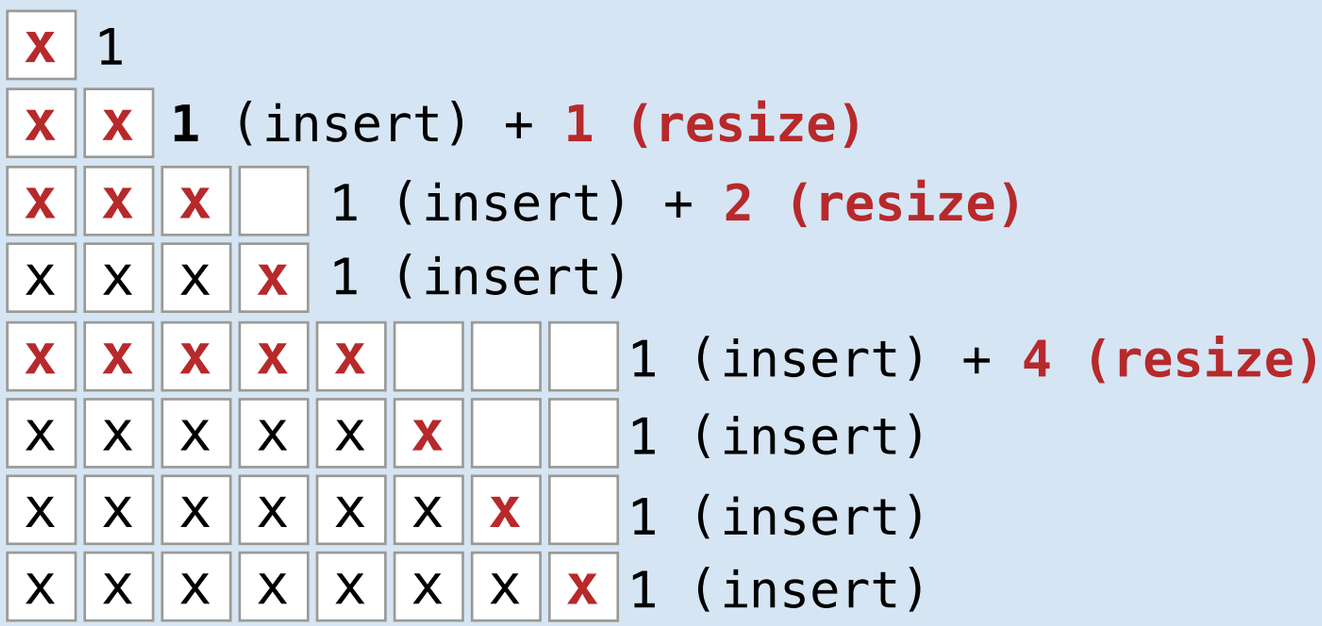

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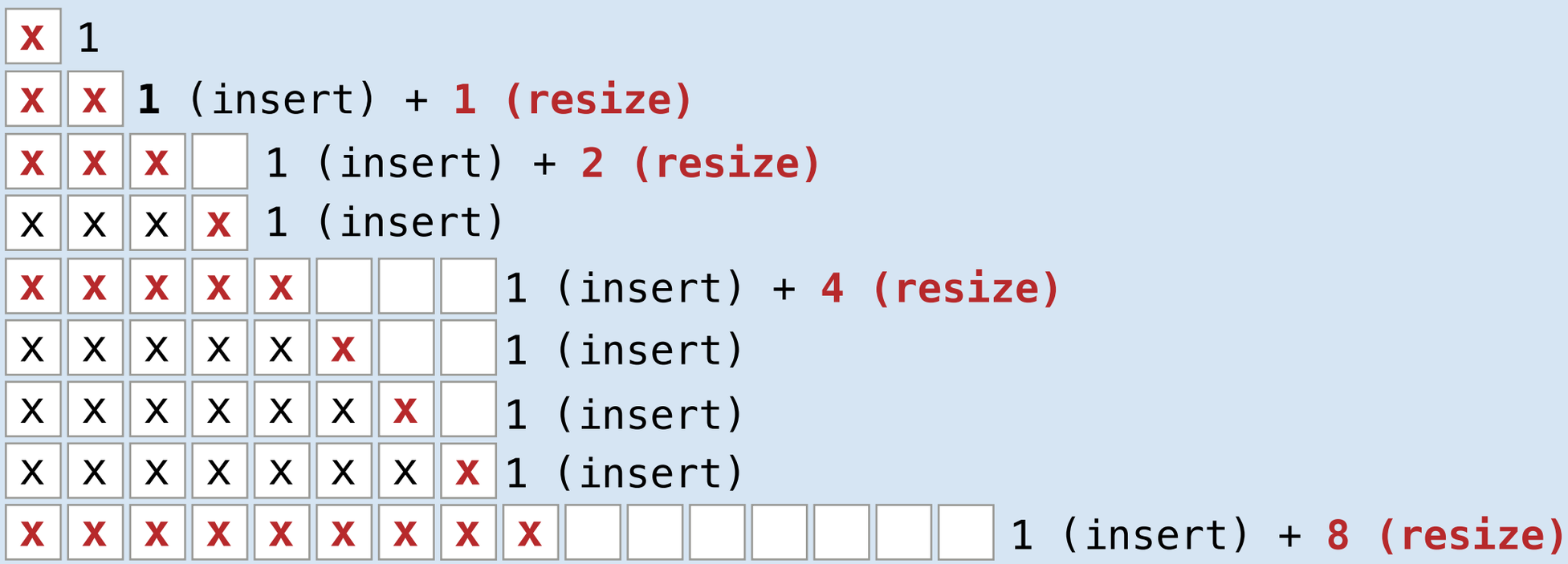

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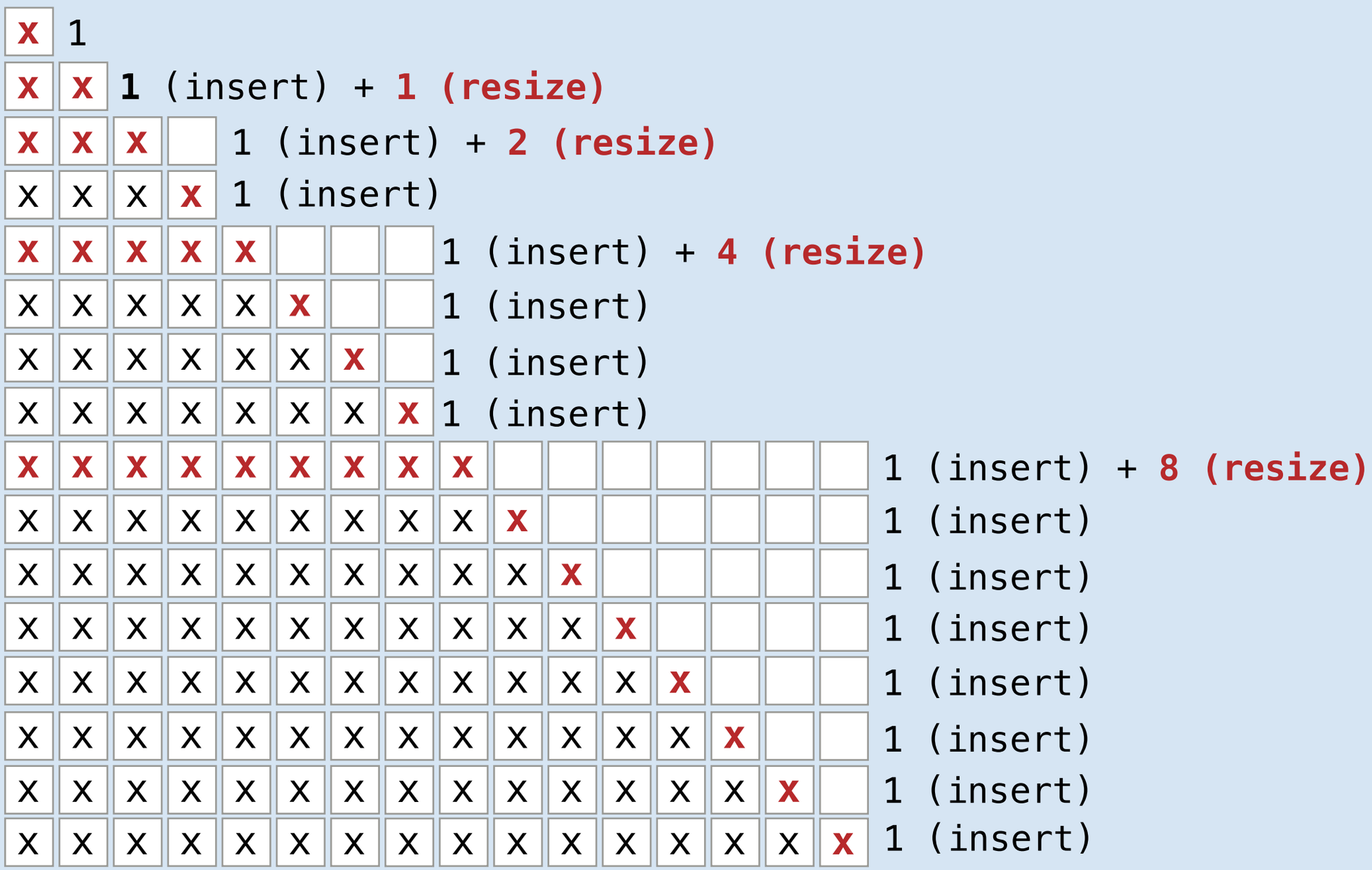

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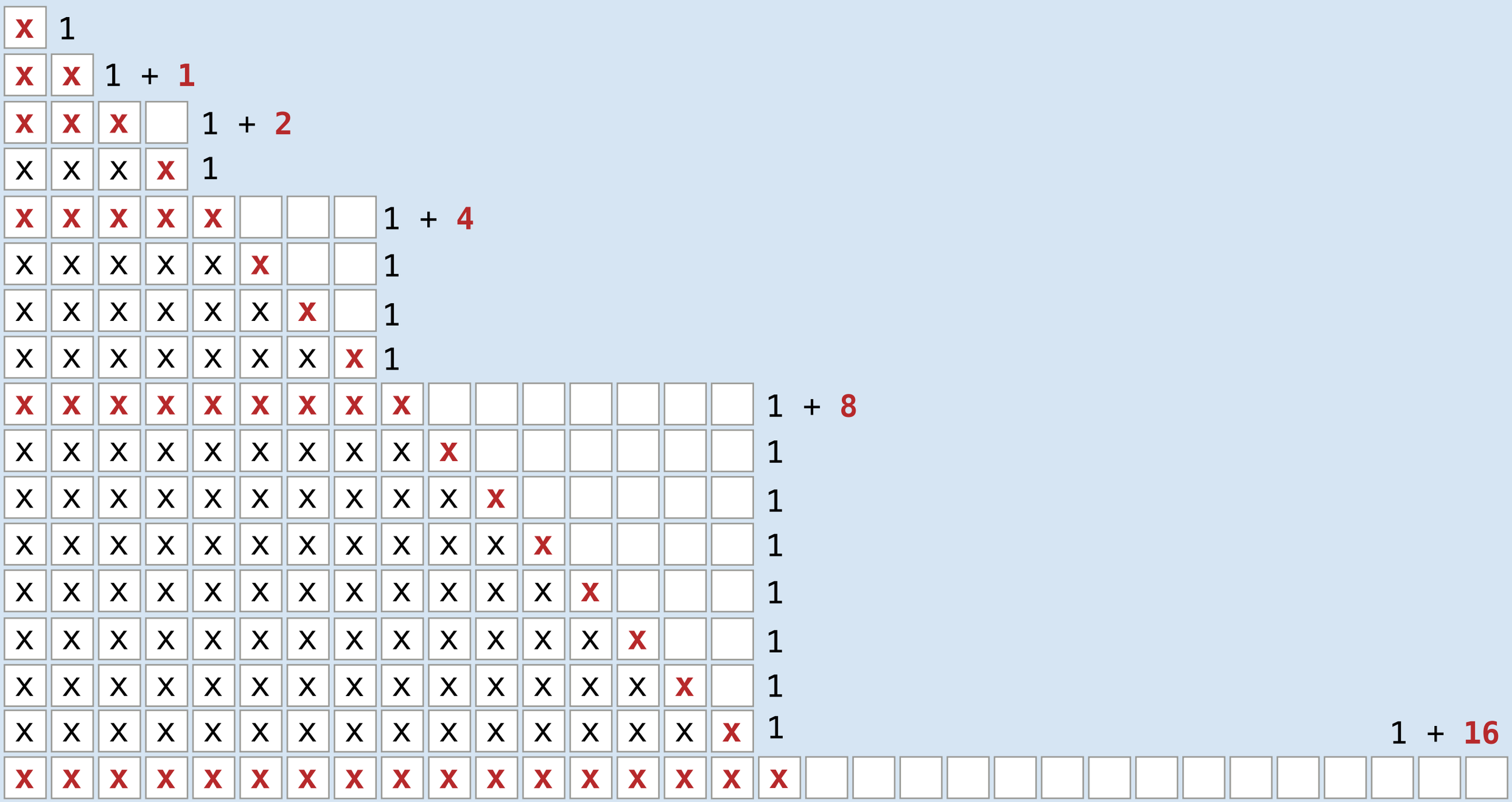

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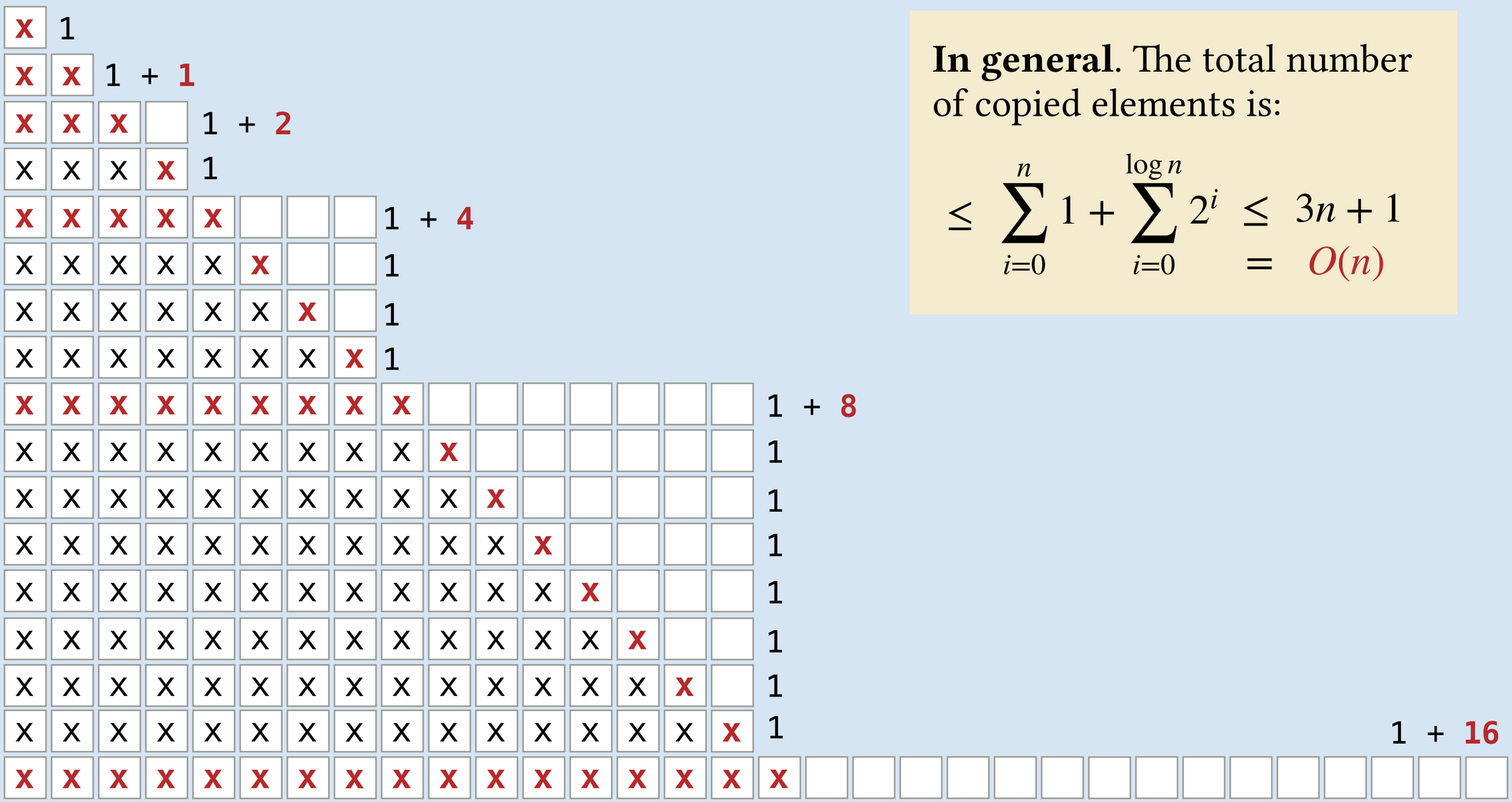

In general. The total number of copied elements is:

$$\leq \sum_{i=0}^{n} 1 + \sum_{i=0}^{\log n} 2^i \leq 3n + 1 = O(n)$$

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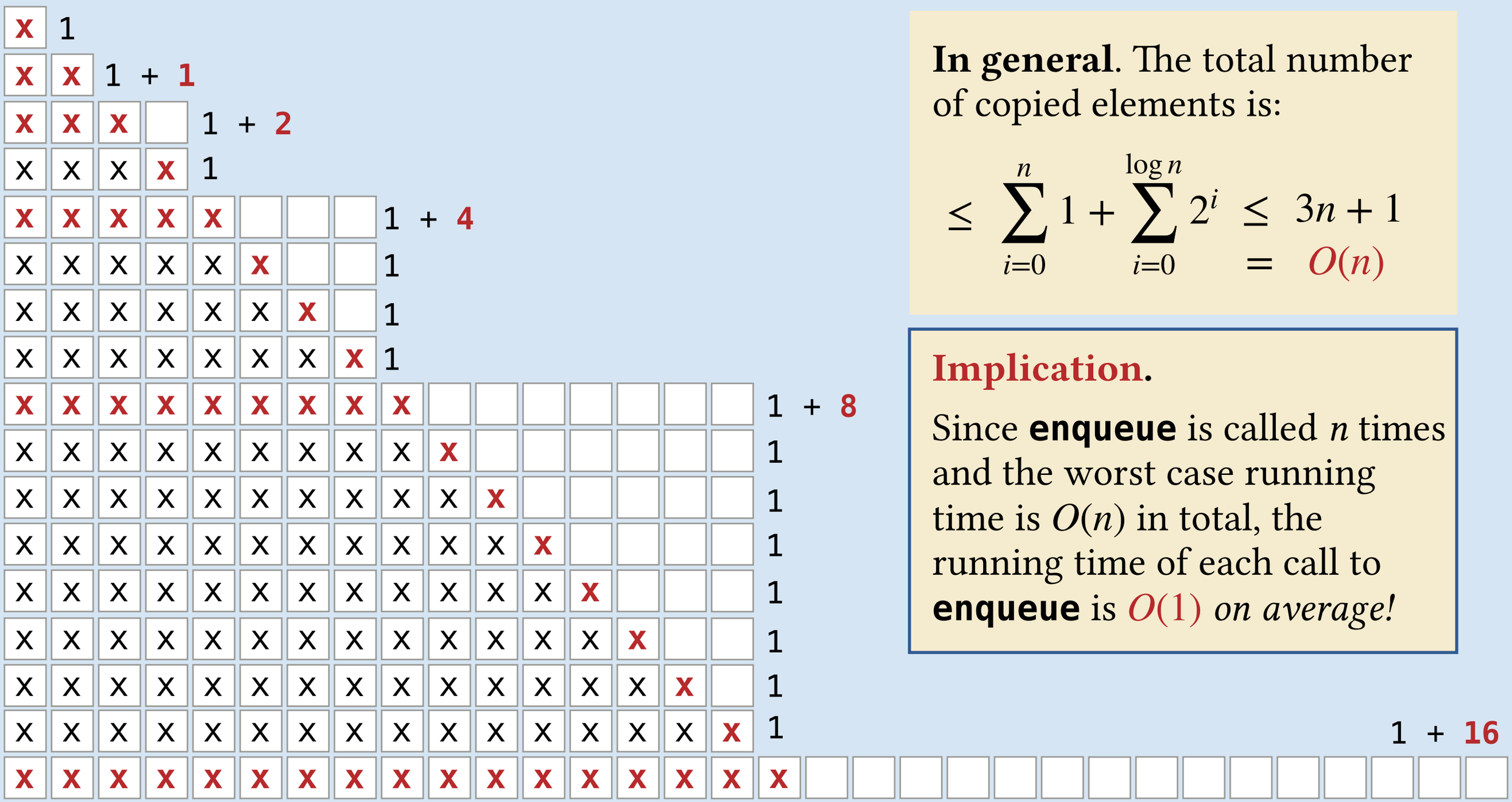

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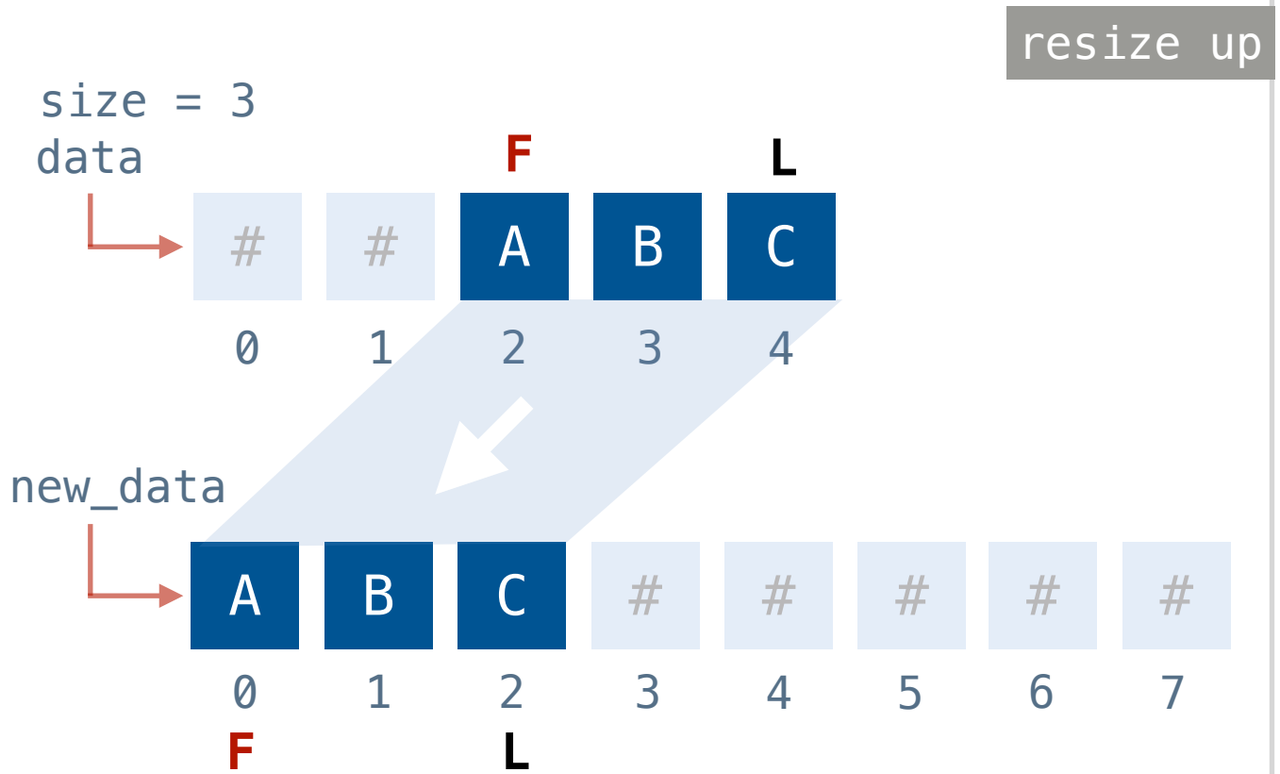
Implication.

Since **enqueue** is called n times and the worst case running time is $O(n)$ in total, the running time of each call to **enqueue** is $O(1)$ *on average!*

Resizing The Queue (resize and reorder the elements)



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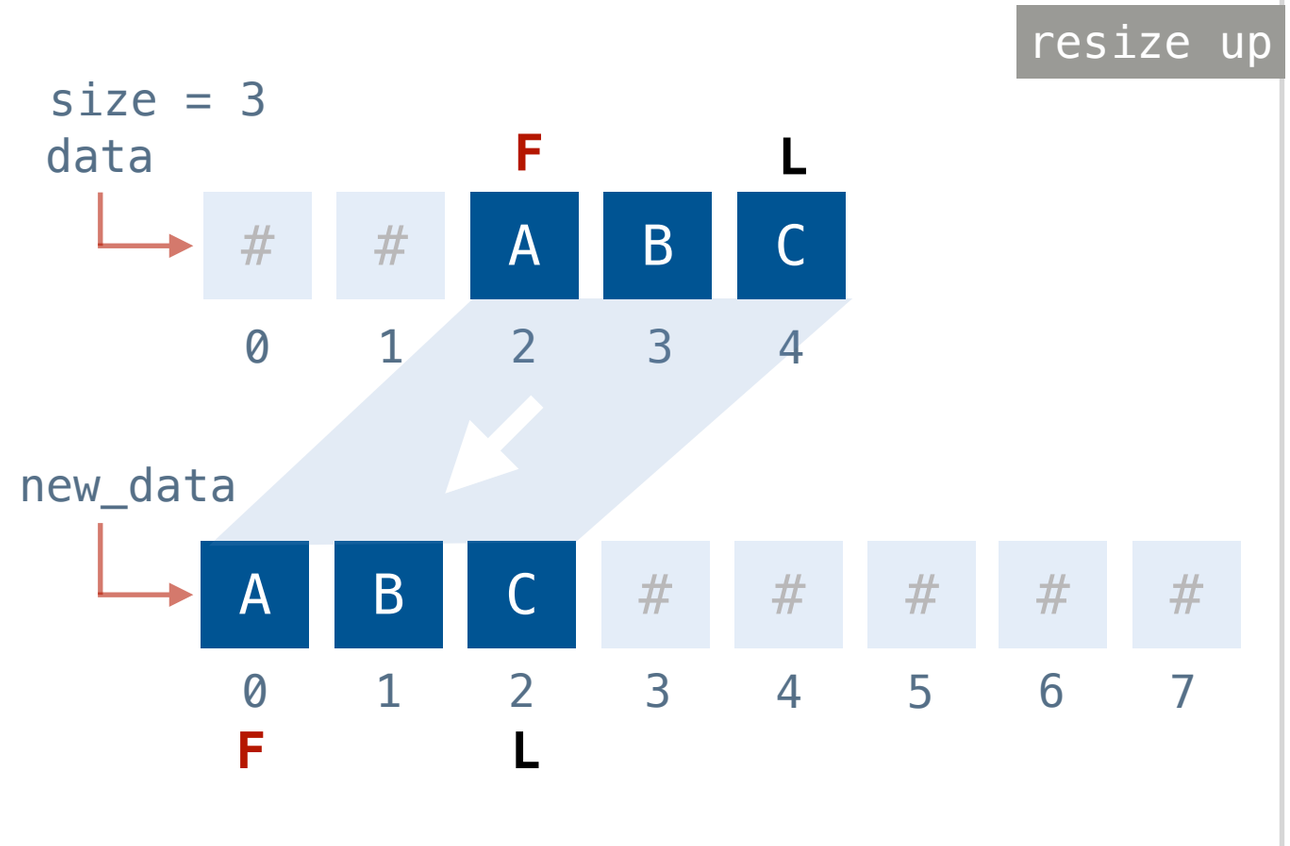

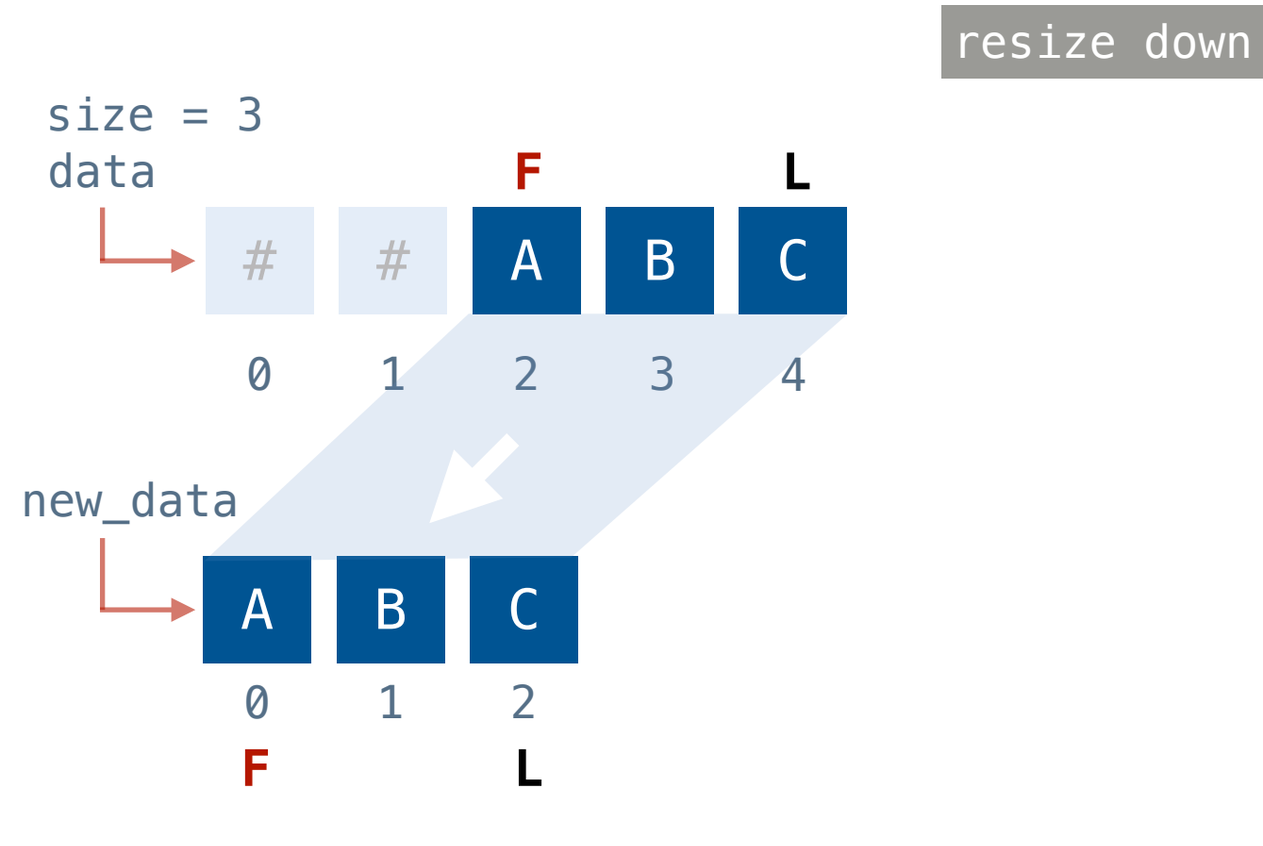

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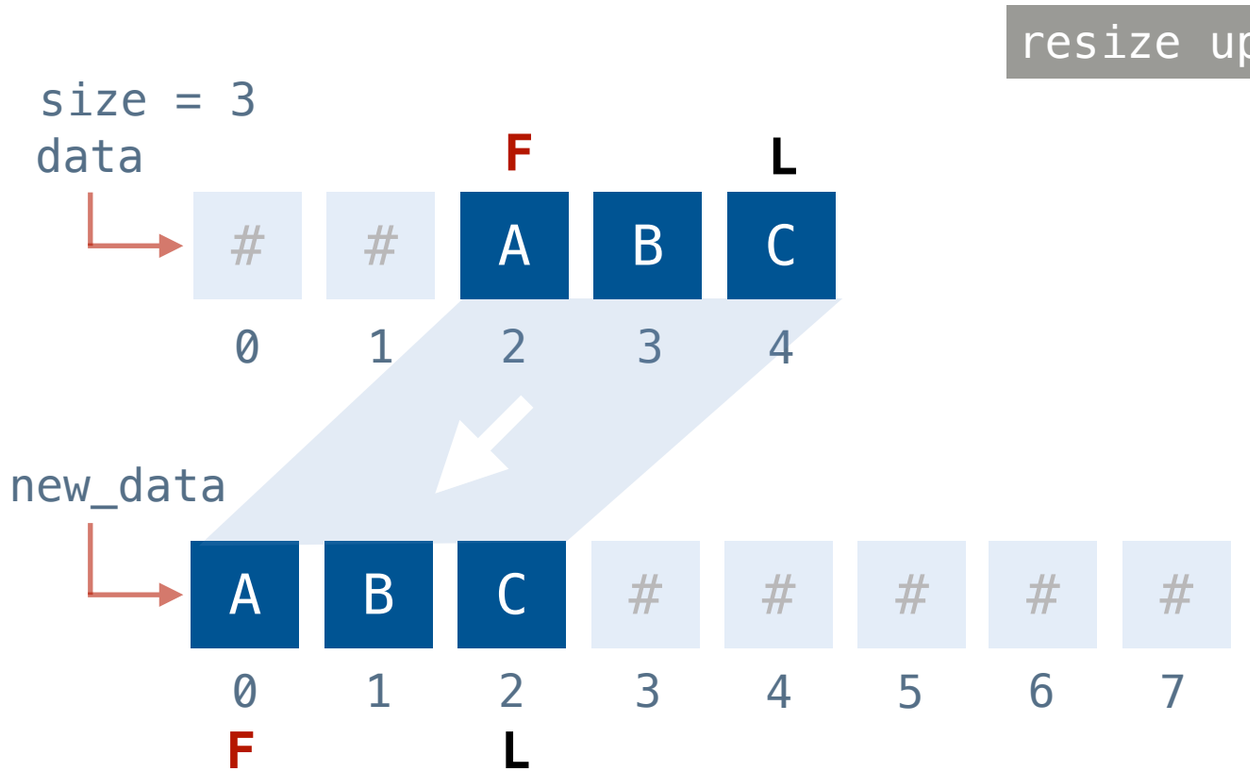

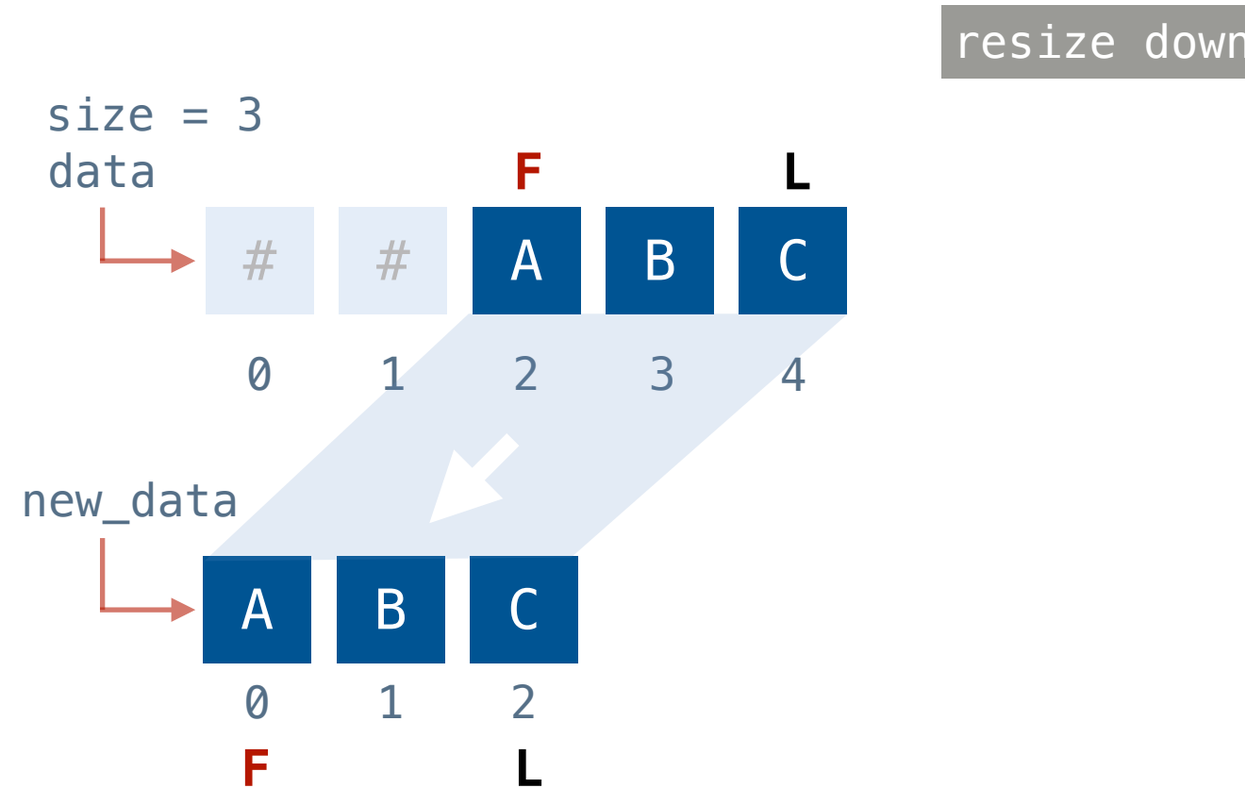

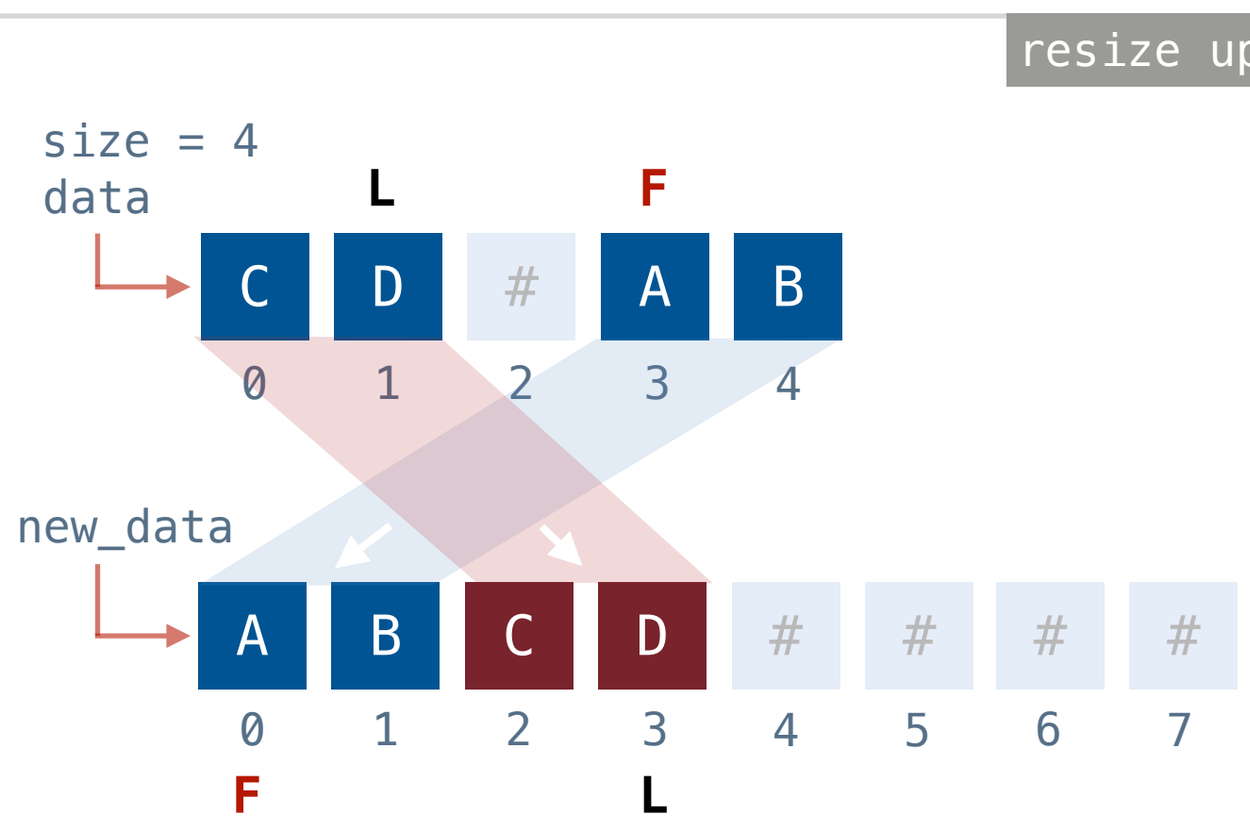

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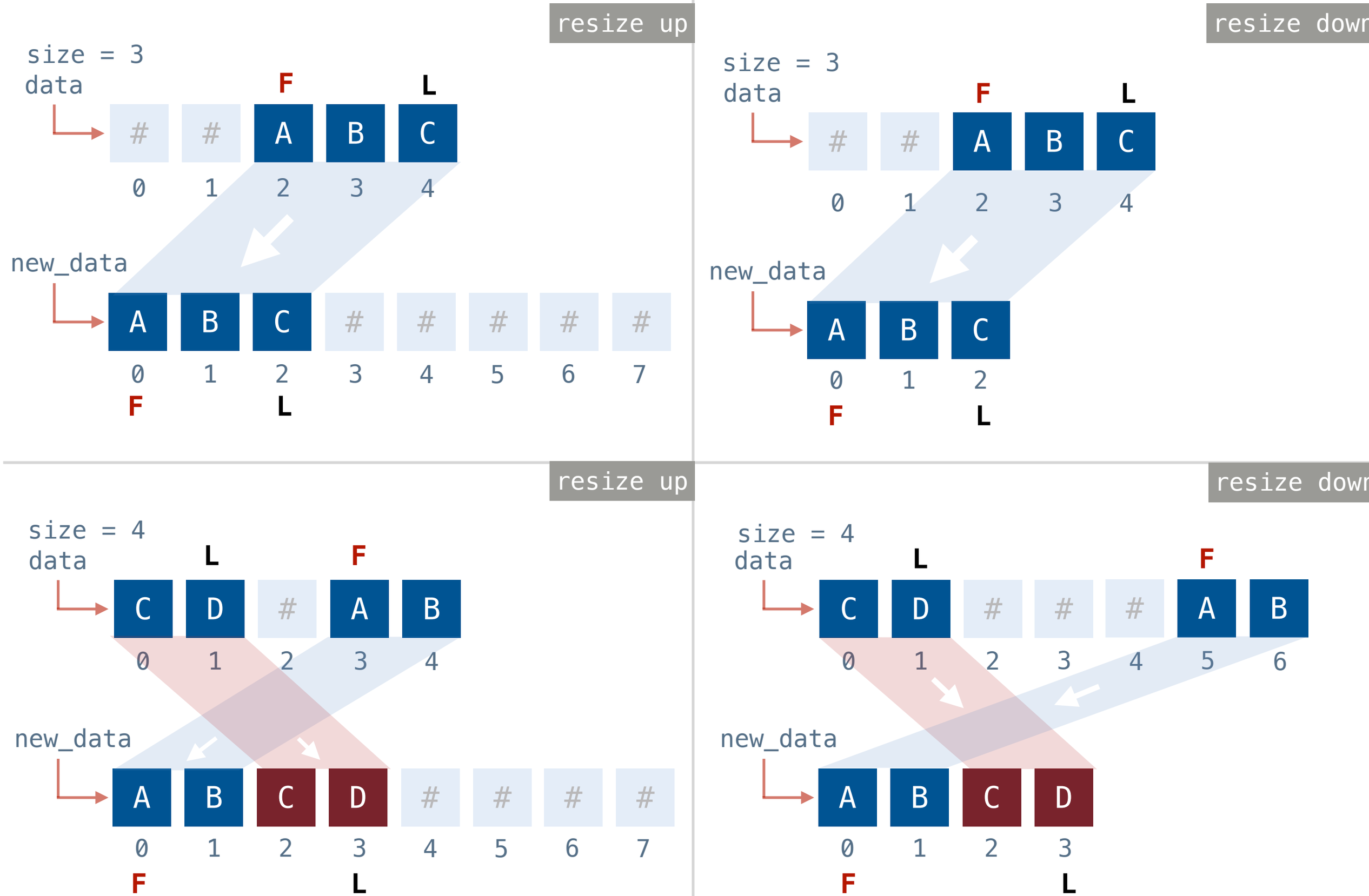

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